68. The value of $\dfrac{\sin 10°+\sin 20°}{\cos 10°+\cos 20°}$ equals

(a) $2+\sqrt{3}$ (b) $\sqrt{2}-1$ (c) $2-\sqrt{3}$ (d) $\sqrt{2}+1$

69. The expression $\cos^6\theta+\sin^6\theta+3\sin^2\theta\cos^2\theta$ simplifies to :

(a) 0 (b) 1 (c) 2 (d) 3

70. $\dfrac{\sin x+\cos x}{\sin x-\cos x}-\dfrac{\sec^2 x+2}{\tan^2 x-1}=$, where $x\in\left(0,\dfrac{\pi}{2}\right)$

(a) $\dfrac{1}{\tan x+1}$ (b) $\dfrac{2}{1+\tan x}$ (c) $\dfrac{2}{1+\cot x}$ (d) $\dfrac{2}{1-\tan x}$

71. If $\dfrac{\cot\alpha+\cot(270°+\alpha)}{\cot\alpha-\cot(270°+\alpha)}-2\cos(135°+\alpha)\cos(315°-\alpha)=\lambda\cos 2\alpha$, where $\alpha\in\left(0,\dfrac{\pi}{2}\right)$, then $\lambda=$

(a) 0 (b) 1 (c) 2 (d) 4

72. The expression $\dfrac{\sin\alpha+\cos\alpha}{\cos\alpha-\sin\alpha}\tan\left(\dfrac{\pi}{4}+\alpha\right)+1$, $\alpha\in\left(-\dfrac{\pi}{4},\dfrac{\pi}{4}\right)$ simplifies to :

(a) $\operatorname{cosec}^2\left(\dfrac{\pi}{4}-\alpha\right)$ (b) $\sec^2\left(\dfrac{\pi}{4}-\alpha\right)$ (c) $\tan^2\left(\dfrac{\pi}{4}-\alpha\right)$ (d) $\cot^2\left(\dfrac{\pi}{4}-\alpha\right)$

73. The value of expression $\dfrac{\tan\alpha+\sin\alpha}{2\cos^2\dfrac{\alpha}{2}}$ for $\alpha=\dfrac{\pi}{4}$ is :

(a) 4 (b) 3 (c) 2 (d) 1

74. $\cos 2\alpha-\cos 3\alpha-\cos 4\alpha+\cos 5\alpha$ simplifies to :

(a) $-4\sin\dfrac{\alpha}{2}\sin\alpha\cos\dfrac{7\alpha}{2}$ (b) $4\sin\dfrac{\alpha}{2}\sin\alpha\cos\dfrac{7\alpha}{2}$

(c) $-4\sin\dfrac{\alpha}{2}\sin\dfrac{7\alpha}{2}\cos\alpha$ (d) $-4\sin\alpha\cos\dfrac{\alpha}{2}\sin\dfrac{7\alpha}{2}$

75. If $\tan\gamma=\sec\alpha\sec\beta+\tan\alpha\tan\beta$, then the least value of $\cos 2\gamma$ is :

(a) -1 (b) $\dfrac{1}{2}$ (c) $-\dfrac{1}{2}$ (d) 0

76. If $\operatorname{cosec} x=\dfrac{2}{\sqrt{3}}$, $\cot x=-\dfrac{1}{\sqrt{3}}$, $x\in[0,2\pi]$, then $\cos x+\cos 2x+\cos 3x+\ldots\ldots+\cos 100x=$

(a) $\dfrac{1}{2}$ (b) $-\dfrac{1}{2}$ (c) $-\dfrac{\sqrt{3}}{2}$ (d) $\dfrac{\sqrt{3}}{2}$

77. The value of $\displaystyle\sum_{r=0}^{10}\cos^3\left(\frac{\pi r}{3}\right)$ is equal to :

(a) $-\dfrac{7}{8}$ (b) $-\dfrac{9}{8}$ (c) $-\dfrac{3}{8}$ (d) $-\dfrac{1}{8}$

78. The value of the expression $\frac{1-4\sin 10°\sin 70°}{2\sin 10°}$ is :

(a) 1 (b) 2 (c) $\sqrt{3}$ (d) $\frac{\sqrt{3}}{2}$

79. If $x, y \in R$ and satisfy $(x+5)^2+(y-12)^2=14^2$, then the minimum value of x^2+y^2 is :

(a) 2 (b) 1 (c) $\sqrt{3}$ (d) $\sqrt{2}$

80. If θ_1, θ_2 and θ_3 are the three values of $\theta \in [0, 2\pi]$ for which $\tan\theta = \lambda$ then the value of $\tan\frac{\theta_1}{3}\tan\frac{\theta_2}{3}+\tan\frac{\theta_2}{3}\tan\frac{\theta_3}{3}+\tan\frac{\theta_3}{3}\tan\frac{\theta_1}{3}$ is equal to (λ is a constant)

(a) -3 (b) -2 (c) 2 (d) 3

81. If $\tan\alpha = \frac{b}{a}$, $a > b > 0$ and if $0 < \alpha < \frac{\pi}{4}$, then $\sqrt{\frac{a+b}{a-b}}+\sqrt{\frac{a-b}{a+b}}$ is equal to :

(a) $\frac{2\sin\alpha}{\sqrt{\cos 2\alpha}}$ (b) $\frac{2\cos\alpha}{\sqrt{\cos 2\alpha}}$ (c) $\frac{2\sin\alpha}{\sqrt{\sin 2\alpha}}$ (d) $\frac{2\cos\alpha}{\sqrt{\sin 2\alpha}}$

82. Minimum value of $3\sin\theta + 4\cos\theta$ in the interval $\left[0, \frac{\pi}{2}\right]$ is :

(a) -5 (b) 3 (c) 4 (d) $\frac{7}{\sqrt{2}}$

83. If $f(n) = \prod_{r=1}^{n} \cos r, n \in N$, then

(a) $|f(n)| > |f(n+1)|$ (b) $f(5) > 0$ (c) $f(4) > 0$ (d) $|f(n)| < |f(n+1)|$

84. If $\tan A + \sin A = p$ and $\tan A - \sin A = q$, then the value of $\frac{(p^2-q^2)^2}{pq}$ is :

(a) 16 (b) 22 (c) 18 (d) 42

85. Let $t_1 = (\sin\alpha)^{\cos\alpha}$, $t_2 = (\sin\alpha)^{\sin\alpha}$, $t_3 = (\cos\alpha)^{\cos\alpha}$, $t_4 = (\cos\alpha)^{\sin\alpha}$, where $\alpha \in \left(0, \frac{\pi}{4}\right)$, then which of the following is correct

(a) $t_3 > t_1 > t_2$ (b) $t_4 > t_2 > t_1$ (c) $t_4 > t_1 > t_2$ (d) $t_1 > t_3 > t_2$

86. If $\cos A = \frac{3}{4}$, then the value of expression $32\sin\frac{A}{2}\sin\frac{5A}{2}$ is equal to :

(a) 11 (b) -11 (c) 12 (d) 4

87. If $\cos(\alpha+\beta)+\sin(\alpha-\beta)=0$ and $\tan\beta = \frac{1}{2009}$; then $\tan 3\alpha$ is :

(a) 2 (b) 1 (c) 3 (d) 4

88. If $2^x = 3^y = 6^{-z}$, the value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is equal to :

(a) 0 (b) 1 (c) 2 (d) 3

89. Let α, β be such that $\pi < \alpha - \beta < 3\pi$

If $\sin\alpha + \sin\beta = -\frac{21}{65}$ and $\cos\alpha + \cos\beta = -\frac{27}{65}$ then the value of $\cos\left(\frac{\alpha-\beta}{2}\right)$ is :

(a) $\frac{-3}{\sqrt{130}}$ (b) $\frac{3}{\sqrt{130}}$ (c) $\frac{6}{65}$ (d) $-\frac{6}{65}$

90. If $\mu = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta} + \sqrt{a^2\sin^2\theta + b^2\cos^2\theta}$ then the difference between maximum and minimum values of μ^2 is :

(a) $2(a^2+b^2)$ (b) $(a+b)^2$ (c) $2\sqrt{a^2+b^2}$ (d) $(a-b)^2$

91. If $P = (\tan(3^{n+1}\theta) - \tan\theta)$ and $Q = \sum_{r=0}^{n} \frac{\sin(3^r\theta)}{\cos(3^{r+1}\theta)}$, then

(a) $P = 2Q$ (b) $P = 3Q$ (c) $2P = Q$ (d) $3P = Q$

92. If $270° < \theta < 360°$, then find $\sqrt{2+\sqrt{2(1+\cos\theta)}}$

(a) $-2\sin\left(\frac{\theta}{4}\right)$ (b) $2\sin\left(\frac{\theta}{4}\right)$ (c) $\pm 2\sin\frac{\theta}{4}$ (d) $2\cos\frac{\theta}{4}$

93. If $y = (\sin x + \cos x) + (\sin 4x + \cos 4x)^2$, then :

(a) $y > 0 \;\forall\; x \in R$ (b) $y \geq 0 \;\forall\; x \in R$

(c) $y < 2+\sqrt{2} \;\forall\; x \in R$ (d) $y = 2+\sqrt{2}$ for some $x \in R$

94. If $\cos x + \cos y + \cos z = \sin x + \sin y + \sin z = 0$ then $\cos(x-y) =$

(a) 0 (b) $-\frac{1}{2}$ (c) 2 (d) 1

95. The exact value of $\operatorname{cosec} 10° + \operatorname{cosec} 50° - \operatorname{cosec} 70°$ is :

(a) 4 (b) 5 (c) 6 (d) 8

96. If $270° < \theta < 360°$, then find $\sqrt{2+\sqrt{2(1+\cos\theta)}}$:

(a) $-2\sin\left(\frac{\theta}{4}\right)$ (b) $2\sin\left(\frac{\theta}{4}\right)$ (c) $\pm 2\sin\frac{\theta}{4}$ (d) $2\cos\frac{\theta}{4}$

Answers

1.	(c)	**2.**	(d)	**3.**	(a)	**4.**	(c)	**5.**	(b)	**6.**	(c)	**7.**	(d)	**8.**	(d)	**9.**	(a)	**10.**	(c)
11.	(c)	**12.**	(b)	**13.**	(b)	**14.**	(d)	**15.**	(b)	**16.**	(d)	**17.**	(c)	**18.**	(a)	**19.**	(d)	**20.**	(a)
21.	(c)	**22.**	(b)	**23.**	(b)	**24.**	(d)	**25.**	(c)	**26.**	(b)	**27.**	(c)	**28.**	(b)	**29.**	(d)	**30.**	(a)
31.	(b)	**32.**	(c)	**33.**	(b)	**34.**	(c)	**35.**	(c)	**36.**	(a)	**37.**	(a)	**38.**	(a)	**39.**	(d)	**40.**	(d)
41.	(b)	**42.**	(b)	**43.**	(c)	**44.**	(d)	**45.**	(d)	**46.**	(b)	**47.**	(b)	**48.**	(c)	**49.**	(d)	**50.**	(a)
51.	(d)	**52.**	(b)	**53.**	(a)	**54.**	(a)	**55.**	(b)	**56.**	(b)	**57.**	(b)	**58.**	(d)	**59.**	(b)	**60.**	(a)
61.	(d)	**62.**	(a)	**63.**	(d)	**64.**	(a)	**65.**	(a)	**66.**	(b)	**67.**	(b)	**68.**	(c)	**69.**	(b)	**70.**	(b)
71.	(c)	**72.**	(a)	**73.**	(d)	**74.**	(a)	**75.**	(d)	**76.**	(b)	**77.**	(d)	**78.**	(a)	**79.**	(b)	**80.**	(a)
81.	(b)	**82.**	(b)	**83.**	(a)	**84.**	(a)	**85.**	(b)	**86.**	(a)	**87.**	(b)	**88.**	(a)	**89.**	(a)	**90.**	(d)
91.	(a)	**92.**	(b)	**93.**	(c)	**94.**	(b)	**95.**	(c)	**96.**	(b)								

Exercise-2 : One or More than One Answer is/are Correct

1. $\cot 12° \cdot \cot 24° \cdot \cot 28° \cdot \cot 32° \cdot \cot 48° \cdot \cot 88° = \ldots\ldots$
(a) $\tan 45°$
(b) 2
(c) $2\tan 15° \cdot \tan 45° \cdot \tan 75°$
(d) $\tan 15° \cdot \tan 45° \cdot \tan 75°$

2. If the equation $\cot^4 x - 2\operatorname{cosec}^2 x + a^2 = 0$ has at least one solution then possible integral values of a can be :
(a) -1
(b) 0
(c) 1
(d) 2

3. Which of the following is/are true ?
(a) $\tan 1 > \tan^{-1} 1$
(b) $\sin 1 > \cos 1$
(c) $\tan 1 < \sin 1$
(d) $\cos(\cos 1) > \dfrac{1}{\sqrt{2}}$

4. Which of the following is/are +ve ?
(a) $\log_{\sin 1} \tan 1$
(b) $\log_{\cos 1}(1 + \tan 3)$
(c) $\log_{\log_{10} 5}(\cos\theta + \sec\theta)$
(d) $\log_{\tan 15°}(2\sin 18°)$

5. If $\sin\alpha + \cos\alpha = \dfrac{\sqrt{3}+1}{2}$, $0 < \alpha < 2\pi$, then possible values $\tan\dfrac{\alpha}{2}$ can take is/are :
(a) $2-\sqrt{3}$
(b) $\dfrac{1}{\sqrt{3}}$
(c) 1
(d) $\sqrt{3}$

6. If $3\sin\beta = \sin(2\alpha + \beta)$, then :
(a) $(\cot\alpha + \cot(\alpha+\beta))(\cot\beta - 3\cot(2\alpha+\beta)) = 6$
(b) $\sin\beta = \cos(\alpha+\beta)\sin\alpha$
(c) $\tan(\alpha+\beta) = 2\tan\alpha$
(d) $2\sin\beta = \sin(\alpha+\beta)\cos\alpha$

7. If $\sin(x + 20°) = 2\sin x \cos 40°$ where $x \in (0, 90°)$, then which of the following hold good ?
(a) $\sec\dfrac{x}{2} = \sqrt{6} - \sqrt{2}$
(b) $\cot\dfrac{x}{2} = 2 + \sqrt{3}$
(c) $\tan 4x = \sqrt{3}$
(d) $\operatorname{cosec} 4x = 2$

8. If $2(\cos(x-y) + \cos(y-z) + \cos(z-x)) = -3$, then :
(a) $\cos x \cos y \cos z = 1$
(b) $\cos x + \cos y + \cos z = 0$
(c) $\sin x + \sin y + \sin z = 1$
(d) $\cos 3x + \cos 3y + \cos 3z = 12\cos x \cos y \cos z$

9. If $0 < x < \dfrac{\pi}{2}$ and $\sin^n x + \cos^n x \geq 1$, then '$n$' may belong to interval :
(a) $[1, 2)$
(b) $[3, 4]$
(c) $(-\infty, 2]$
(d) $[-1, 1]$

10. If $x = \sin(\alpha-\beta)\cdot\sin(\gamma-\delta)$, $y = \sin(\beta-\gamma)\cdot\sin(\alpha-\delta)$, $z = \sin(\gamma-\alpha)\cdot\sin(\beta-\delta)$, then :
(a) $x + y + z = 0$
(b) $x^3 + y^3 + z^3 = 3xyz$
(c) $x + y - z = 0$
(d) $x^3 + y^3 - z^3 = 3xyz$

11. If $X = x\cos\theta - y\sin\theta, Y = x\sin\theta + y\cos\theta$ and $X^2 + 4XY + Y^2 = Ax^2 + By^2, 0 \le \theta \le \pi/2$, then : (where A and B are constants)

(a) $\theta = \frac{\pi}{6}$ (b) $\theta = \frac{\pi}{4}$ (c) $A = 3$ (d) $B = -1$

12. If $2a = 2\tan 10° + \tan 50°$; $2b = \tan 20° + \tan 50°$
$2c = 2\tan 10° + \tan 70°$; $2d = \tan 20° + \tan 70°$
Then which of the following is/are correct ?

(a) $a + d = b + c$ (b) $a + b = c$ (c) $a > b < c > d$ (d) $a < b < c < d$

13. Which of the following real numbers when simplified are neither terminating nor repeating decimal ?

(a) $\sin 75° \cdot \cos 75°$ (b) $\log_2 28$ (c) $\log_3 5 \cdot \log_5 6$ (d) $8^{-(\log_{27} 3)}$

14. If $\alpha = \sin x \cos^3 x$ and $\beta = \cos x \sin^3 x$, then :

(a) $\alpha - \beta > 0$; for all x in $\left(0, \frac{\pi}{4}\right)$ (b) $\alpha - \beta < 0$; for all x in $\left(0, \frac{\pi}{4}\right)$

(c) $\alpha + \beta > 0$; for all x in $\left(0, \frac{\pi}{2}\right)$ (d) $\alpha + \beta < 0$; for all x in $\left(0, \frac{\pi}{2}\right)$

15. If $\frac{\pi}{2} < \theta < \pi$, then possible answers of $\sqrt{2 + \sqrt{2 + 2\cos 4\theta}}$ is/are :

(a) $2\cos\theta$ (b) $2\sin\theta$ (c) $-2\sin\theta$ (d) $-2\cos\theta$

16. If $\cot^3\alpha + \cot^2\alpha + \cot\alpha = 1$ then which of the following is/are correct:

(a) $\cos 2\alpha \tan\alpha = 1$ (b) $\cos 2\alpha \cdot \tan\alpha = -1$

(c) $\cos 2\alpha - \tan 2\alpha = -1$ (d) $\cos 2\alpha - \tan 2\alpha = 1$

17. All values of $x \in \left(0, \frac{\pi}{2}\right)$ such that $\frac{\sqrt{3} - 1}{\sin x} + \frac{\sqrt{3} + 1}{\cos x} = 4\sqrt{2}$ are :

(a) $\frac{\pi}{15}$ (b) $\frac{\pi}{12}$ (c) $\frac{11\pi}{36}$ (d) $\frac{3\pi}{10}$

18. If $\alpha > \frac{1}{\sin^6 x + \cos^6 x}\ \forall x \in R$, then α can be :

(a) 3 (b) 4 (c) 5 (d) 6

19. If $x \in \left(0, \frac{\pi}{2}\right)$ and $\sin x = \frac{3}{\sqrt{10}}$;

Let $k = \log_{10} \sin x + \log_{10} \cos x + 2\log_{10} \cot x + \log_{10} \tan x$ then the value of k satisfies

(a) $k = 0$ (b) $k + 1 = 0$ (c) $k - 1 = 0$ (d) $k^2 - 1 = 0$

20. If A, B, C are angles of a triangle ABC and $\tan A \tan C = 3$; $\tan B \tan C = 6$ then which is(are) correct :

(a) $A = \frac{\pi}{4}$ (b) $\tan A \tan B = 2$ (c) $\frac{\tan A}{\tan C} = 3$ (d) $\tan B = 2\tan A$

21. The value of $\dfrac{\sin x - \cos x}{\sin^3 x}$ is equal to :

(a) $\operatorname{cosec}^2 x(1-\cot x)$ (b) $1-\cot x+\cot^2 x-\cot^3 x$

(c) $\operatorname{cosec}^2 x-\cot x-\cot^3 x$ (d) $\dfrac{1-\cot x}{\sin^2 x}$

22. If $f(x)=\sin^2 x+\sin^2\left(x+\dfrac{2\pi}{3}\right)+\sin^2\left(x+\dfrac{4\pi}{3}\right)$ then :

(a) $f\left(\dfrac{\pi}{15}\right)=\dfrac{3}{2}$ (b) $f\left(\dfrac{15}{\pi}\right)=\dfrac{2}{3}$ (c) $f\left(\dfrac{\pi}{10}\right)=\dfrac{3}{2}$ (d) $f\left(\dfrac{10}{\pi}\right)=\dfrac{2}{3}$

23. The range of $y=\dfrac{\sin 4x-\sin 2x}{\sin 4x+\sin 2x}$ satisfies

(a) $y\in\left(-\infty,\dfrac{1}{3}\right)$ (b) $y\in\left(\dfrac{1}{3},1\right)$ (c) $y\in(1,3)$ (d) $y\in(3,\infty)$

24. If $\sqrt{2}\cos A=\cos B+\cos^3 B$ and $\sqrt{2}\sin A=\sin B-\sin^3 B$, then the possible value of $\sin(A-B)$ is/are

(a) $\dfrac{1}{2}$ (b) $\dfrac{1}{3}$ (c) $-\dfrac{1}{2}$ (d) $-\dfrac{1}{3}$

25. If $\alpha>\dfrac{1}{\sin^6 x+\cos^6 x}\ \forall\, x\in R$, then α can be

(a) 3 (b) 4 (c) 5 (d) 6

26. If $\cot^3\alpha+\cot^2\alpha+\cot\alpha=1$ then which of the following is/are correct

(a) $\cos 2\alpha\tan\alpha=1$ (b) $\cos 2\alpha\cdot\tan\alpha=-1$

(c) $\cos 2\alpha-\tan 2\alpha=-1$ (d) $\cos 2\alpha-\tan 2\alpha=1$

Answers

1.	(a, d)	**2.**	(a, b, c)	**3.**	(a, b, d)	**4.**	(b, d)	**5.**	(a, b)	**6.**	(a, b, c, d)
7.	(a, b)	**8.**	(b, d)	**9.**	(a, c, d)	**10.**	(a, b)	**11.**	(b, c, d)	**12.**	(a, b, d)
13.	(b, c)	**14.**	(a, c)	**15.**	(b, d)	**16.**	(b, d)	**17.**	(b, c)	**18.**	(b, d)
19.	(b, d)	**20.**	(a, b, d)	**21.**	(a, b, c, d)	**22.**	(a, c)	**23.**	(a, d)	**24.**	(b, d)
25.	(c, d)	**26.**	(b, d)								

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

Let $l = \sin\theta, m = \cos\theta$ and $n = \tan\theta$.

1. If $\theta = 5$ radian, then :

(a) $l > m$ (b) $l < m$ (c) $l = m$ (d) none of these

2. If $\theta = -1042°$, then :

(a) $n > 1$ (b) $n < 1$ (c) $n = 1$ (d) nothing can be said

3. If $\theta = 7$ radian, then :

(a) $l + m > 0$ (b) $l + m < 0$ (c) $l + m = 0$ (d) nothing can be said

Paragraph for Question Nos. 4 to 6

Let a, b, c are respectively the sines and p, q, r are respectively the cosines of $\alpha, \alpha + \frac{2\pi}{3}$ and $\alpha + \frac{4\pi}{3}$, then :

4. The value of $(a + b + c)$ is :

(a) 0 (b) $\frac{3}{4}$ (c) 1 (d) none of these

5. The value of $(ab + bc + ca)$ is :

(a) 0 (b) $-\frac{3}{4}$ (c) $-\frac{1}{2}$ (d) -1

6. The value of $(qc - rb)$ is :

(a) 0 (b) $-\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) depends on α

Paragraph for Question Nos. 7 to 8

Consider a right angle triangle ABC right angle at B such that $AC = \sqrt{8 + 4\sqrt{3}}$ and $AB = 1$. A line through vertex A meet BC at D such that $AB = BD$. An arc DE of radius AD is drawn from vertex A to meet AC at E and another arc DF of radius CD is drawn from vertex C to meet AC at F. On the basis of above information, answer the following questions.

7. $\sqrt{\tan A + \cot C}$ is equal to :

(a) $\sqrt{3}$ (b) 1 (c) $2 + \sqrt{3}$ (d) $\sqrt{3} + 1$

8. $\log_{AE}\left(\frac{AC}{CD}\right)$ is equal to :

(a) $\sqrt{2}$ (b) 1 (c) 0 (d) -1

Paragraph for Question Nos. 9 to 10

Consider a triangle ABC such that $\cot A + \cot B + \cot C = \cot\theta$. Now answer the following :

9. The possible value of θ is :

(a) $60°$ (b) $25°$ (c) $35°$ (d) $45°$

10. $\sin(A-\theta)\sin(B-\theta)\sin(C-\theta) =$:

(a) $\tan^3\theta$ (b) $\cot^3\theta$ (c) $\sin^3\theta$ (d) $\cos^3\theta$

Paragraph for Question Nos. 11 to 12

Consider the function $f(x) = \frac{\sqrt{1+\cos x}+\sqrt{1-\cos x}}{\sqrt{1+\cos x}-\sqrt{1-\cos x}}$ then

11. If $x \in (\pi, 2\pi)$ then $f(x)$ is :

(a) $\cot\left(\frac{\pi}{2}+\frac{x}{2}\right)$ (b) $\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)$ (c) $\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)$ (d) $\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)$

12. If the value of $f\left(\frac{\pi}{3}\right) = a + b\sqrt{c}$ where $a, b, c \in N$ then the value of $a+b+c$ is :

(a) 4 (b) 5 (c) 6 (d) 7

Answers

1.	(b)	**2.**	(b)	**3.**	(a)	**4.**	(a)	**5.**	(b)	**6.**	(c)	**7.**	(d)	**8.**	(b)	**9.**	(b)	**10.**	(c)
11.	(d)	**12.**	(c)																

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	If $(1+\tan 5^\circ)(1+\tan 10^\circ)\ldots\ldots(1+\tan 45^\circ)=2^{k+1}$ then 'k' equals	**(P)**	0
(B)	Sum of positive integral values of 'a' for which $a^2-6\sin x-5a\le 0\ \forall\ x\in R$ is	**(Q)**	2
(C)	The minimum value of $\dfrac{\left(a+\frac{1}{a}\right)^4-\left(a^4+\frac{1}{a^4}\right)-2}{\left(a+\frac{1}{a}\right)^2+a^2+\frac{1}{a^2}}$ is	**(R)**	5
(D)	Number of real roots of the equation $\sum_{k=1}^{3}(x-k)^2=0$ is	**(S)**	4
		(T)	5

2.

	Column-I		Column-II
(A)	Maximum value of $y=\dfrac{1-\tan^2(\pi/4-x)}{1+\tan^2(\pi/4-x)}$	**(P)**	1
(B)	Minimum value of $\log_3\left(\dfrac{5\sin x-12\cos x+26}{13}\right)$	**(Q)**	0
(C)	Minimum value of $y=-2\sin^2 x+\cos x+3$	**(R)**	$\frac{7}{8}$
(D)	Maximum value of $y=4\sin^2\theta+4\sin\theta\cos\theta+\cos^2\theta$	**(S)**	5
		(T)	6

3.

	Column-I		Column-II
(A)	The value of $\dfrac{\cos 68^\circ}{\sin 56^\circ \sin 34^\circ \tan 22^\circ}$ equals to	**(P)**	16
(B)	The value of $(\cos 65^\circ+\sqrt{3}\sin 5^\circ+\cos 5^\circ)^2=\lambda\cos^2 25^\circ$; then value of λ be	**(Q)**	3

(C)	If $\cos A = \frac{3}{4}$; then the value of $\frac{32}{11}\sin\frac{A}{2}\sin\frac{5A}{2}$ is equal to	**(R)**	4
(D)	If $7\log_a\frac{16}{15} + 5\log_a\frac{25}{24} + 3\log_a\frac{81}{80} = 8$ then the value of a^{16} equals to	**(S)**	2
		(T)	1

4.

	Column-I		Column-II
(A)	If $\sin x + \cos x = \frac{1}{5}$; then $\lvert 12\tan x\rvert$ is equal to	**(P)**	2
(B)	Number of values of θ lying in $(-2\pi, \pi)$ and satisfying $\cot\frac{\theta}{2} = (1 + \cot\theta)$ is	**(Q)**	6
(C)	If $2 - \sin^4 x + 8\sin^2 x = \alpha$ has solution, then α can be	**(R)**	9
(D)	Number of integral values of x satisfying $\log_4(2x^2 + 5x + 27) - \log_2(2x - 1) \geq 0$	**(S)**	14
		(T)	16

5. Match the function given in **Column-I** to the number of integers in its range given in **Column-II.**

	Column-I		Column-II
(A)	$f(x) = 2\cos^2 x + \sin x - 8$	**(P)**	5
(B)	$f(x) = \sin^2 x + 3\cos^2 x + 5$	**(Q)**	4
(C)	$f(x) = 4\sin x\cos x - \sin^2 x + 3\cos^2 x$	**(R)**	3
(D)	$f(x) = \cos(\sin x) + \sin(\sin x)$	**(S)**	2

Answers

1. $A \to S$; $B \to R$; $C \to Q$; $D \to P$

2. $A \to P$; $B \to Q$; $C \to R$; $D \to S$

3. $A \to S$; $B \to Q$; $C \to T$; $D \to R$

4. $A \to R, T$; $B \to P$; $C \to P, Q, R$; $D \to Q$

5. $A \to Q$; $B \to R$; $C \to P$; $D \to S$

Exercise-5 : Subjective Type Problems

1. Let $P = \dfrac{\sin 80^\circ \sin 65^\circ \sin 35^\circ}{\sin 20^\circ + \sin 50^\circ + \sin 110^\circ}$, then the value of $24P$ is :

2. The value of expression $(1 - \cot 23^\circ)(1 - \cot 22^\circ)$ is equal to :

3. If $\tan A$ and $\tan B$ are the roots of the quadratic equation, $4x^2 - 7x + 1 = 0$ then evaluate $4\sin^2(A+B) - 7\sin(A+B)\cdot\cos(A+B) + \cos^2(A+B)$.

4. $A_1A_2A_3 \ldots\ldots A_{18}$ is a regular 18 sided polygon. B is an external point such that A_1A_2B is an equilateral triangle. If $A_{18}A_1$ and A_1B are adjacent sides of a regular n sided polygon, then $n =$

5. If $10\sin^4\alpha + 15\cos^4\alpha = 6$ and the value of $9\operatorname{cosec}^4\alpha + \beta\sec^4\alpha$ is S, then find the value of $\dfrac{S}{25}$.

6. The value of $\left(1 + \tan\dfrac{3\pi}{8}\tan\dfrac{\pi}{8}\right) + \left(1 + \tan\dfrac{5\pi}{8}\tan\dfrac{3\pi}{8}\right) + \left(1 + \tan\dfrac{7\pi}{8}\tan\dfrac{5\pi}{8}\right) + \left(1 + \tan\dfrac{9\pi}{8}\tan\dfrac{7\pi}{8}\right)$

7. If $\alpha = \dfrac{\pi}{7}$ then find the value of $\left(\dfrac{1}{\cos\alpha} + \dfrac{2\cos\alpha}{\cos 2\alpha}\right)$.

8. Given that for $a, b, c, d \in R$, if $a\sec(200^\circ) - c\tan(200^\circ) = d$ and $b\sec(200^\circ) + d\tan(200^\circ) = c$, then find the value of $\left(\dfrac{a^2 + b^2 + c^2 + d^2}{bd - ac}\right)\sin 20^\circ$.

9. The expression $2\cos\dfrac{\pi}{17}\cdot\cos\dfrac{9\pi}{17} + \cos\dfrac{7\pi}{17} + \cos\dfrac{9\pi}{17}$ simplifies to an integer P. Find the value of P.

10. If the expression $\dfrac{\sin\theta\sin 2\theta + \sin 3\theta\sin 6\theta + \sin 4\theta\sin 13\theta}{\sin\theta\cos 2\theta + \sin 3\theta\cos 6\theta + \sin 4\theta\cos 13\theta} = \tan k\theta$, where $k \in N$. Find the value of k.

11. Let $a = \sin 10^\circ, b = \sin 50^\circ, c = \sin 70^\circ$, then $8abc\left(\dfrac{a+b}{c}\right)\left(\dfrac{1}{a} + \dfrac{1}{b} - \dfrac{1}{c}\right)$ is equal to

12. If $\sin^3\theta + \sin^3\left(\theta + \dfrac{2\pi}{3}\right) + \sin^3\left(\theta + \dfrac{4\pi}{3}\right) = a\sin b\theta$. Find the value of $\left|\dfrac{b}{a}\right|$.

13. If $\displaystyle\sum_{r=1}^{n}\left(\dfrac{\tan 2^{r-1}}{\cos 2^r}\right) = \tan p^n - \tan q$, then find the value of $(p+q)$.

14. If $x = \sec\theta - \tan\theta$ and $y = \operatorname{cosec}\theta + \cot\theta$, then $y - x - xy =$

15. If $\cos 18^\circ - \sin 18^\circ = \sqrt{n}\sin 27^\circ$, then $n =$

16. The value of $3(\sin 1 - \cos 1)^4 + 6(\sin 1 + \cos 1)^2 + 4(\sin^6 1 + \cos^6 1)$ is equal to

17. If $x = \alpha$ satisfy the equation $3^{\sin 2x + 2\cos^2 x} + 3^{1 - \sin 2x + 2\sin^2 x} = 28$, then $(\sin 2\alpha - \cos 2\alpha)^2 + 8\sin 4\alpha$ is equal to :

18. The least value of the expression $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 \;\forall\, \theta \in R$ is

19. If $\tan 20^\circ + \tan 40^\circ + \tan 80^\circ - \tan 60^\circ = \lambda\sin 40^\circ$, then λ is equal to

20. If $K°$ lies between 360° and 540° and $K°$ satisfies the equation

$1 + \cos 10x \cos 6x = 2\cos^2 8x + \sin^2 8x$, then $\frac{K}{10} =$

21. If $\cos 20° + 2\sin^2 55° = 1 + \sqrt{2} \sin K°, K \in (0, 90)$, then $K =$

22. The exact value of $\text{cosec}\,10° + \text{cosec}\,50° - \text{cosec}\,70°$ is :

23. Let α be the smallest integral value of $x, x > 0$ such that $\tan 19x = \frac{\cos 96° + \sin 96°}{\cos 96° - \sin 96°}$. The last digit of α is :

24. Find the value of the expression $\frac{\sin 20°(4\cos 20° + 1)}{\cos 20° \cos 30°}$

25. If the value of $\cos\frac{2\pi}{7} + \cos\frac{4\pi}{7} + \cos\frac{6\pi}{7} + \cos\frac{7\pi}{7} = -\frac{l}{2}$. Find the value of l.

26. If $\cos A = \frac{3}{4}$ and $k \sin\left(\frac{A}{2}\right)\sin\left(\frac{5A}{2}\right) = \frac{11}{8}$. Find k.

27. Find the least value of the expression $3\sin^2 x + 4\cos^2 x$.

28. If $\tan\alpha$ and $\tan\beta$ are the roots of equation $x^2 - 12x - 3 = 0$, then the value of $\sin^2(\alpha+\beta) + 2\sin(\alpha+\beta)\cos(\alpha+\beta) + 5\cos^2(\alpha+\beta)$ is :

29. The value of $\frac{\cos 24°}{2\tan 33° \sin^2 57°} + \frac{\sin 162°}{\sin 18° - \cos 18° \tan 9°} + \cos 162°$ is equal to :

30. Find the value of $\tan\theta(1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta)$, when $\theta = \frac{\pi}{32}$.

31. If λ be the minimum value of $y = (\sin x + \text{cosex}\, x)^2 + (\cos x + \sec x)^2 + (\tan x + \cot x)^2$ where $x \in R$. Find $\lambda - 6$.

Answers

1.	6	**2.**	2	**3.**	1	**4.**	9	**5.**	3	**6.**	0	**7.**	4
8.	2	**9.**	0	**10.**	9	**11.**	6	**12.**	4	**13.**	3	**14.**	1
15.	2	**16.**	13	**17.**	1	**18.**	9	**19.**	8	**20.**	45	**21.**	65
22.	6	**23.**	9	**24.**	2	**25.**	3	**26.**	4	**27.**	3	**28.**	2
29.	2	**30.**	1	**31.**	7								

□□□

Chapter 23 – Trigonometric Equations

23 TRIGONOMETRIC EQUATIONS

Exercise-1 : Single Choice Problems

1. Let x and y be 2 real numbers which satisfy the equations $(\tan^2 x - \sec^2 y) = \frac{5a}{6} - 3$ and $(-\sec^2 x + \tan^2 y) = a^2$, then the product of all possible value's of a can be equal to :

(a) 0 (b) $\frac{-2}{3}$ (c) -1 (d) $\frac{-3}{2}$

2. The general solution of the equation $\tan^2(x+y) + \cot^2(x+y) = 1 - 2x - x^2$ lie on the line is :

(a) $x = -1$ (b) $x = -2$ (c) $y = -1$ (d) $y = -2$

3. General solution of the equation $\sin x + \cos x = \min\limits_{a \in R}\{1, a^2 - 4a + 6\}$ is :

(a) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{4}$ (b) $2n\pi + (-1)^n \frac{\pi}{4}$

(c) $n\pi + (-1)^{n+1} \frac{\pi}{4}$ (d) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$

(where $n \in I$, I represent set of integers)

4. The number of solutions of the equation $\left(2\sin\left(\frac{\sin x}{2}\right)\right)\left(\cos\left(\frac{\sin x}{2}\right)\right)\left(\sin\left(2\tan\frac{x}{2}\cos^2\frac{x}{2}\right) - 3\right) + 2 = 0$ in $[0, 2\pi]$ is :

(a) 0 (b) 1 (c) 2 (d) 4

5. Number of solution of $\tan(2x) = \tan(6x)$ in $(0, 3\pi)$ is :

(a) 4 (b) 5 (c) 3 (d) None of these

6. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is :

(a) 0 (b) 2 (c) 6 (d) 8

7. The number of different values of θ satisfying the equation $\cos\theta + \cos 2\theta = -1$, and at the same time satisfying the condition $0 < \theta < 360°$ is :

(a) 1 (b) 2 (c) 3 (d) 4

8. The total number of solutions of the equation $\max(\sin x, \cos x) = \frac{1}{2}$ for $x \in (-2\pi, 5\pi)$ is equal to:

(a) 3 (b) 6 (c) 7 (d) 8

9. The general value of x satisfying the equation $2\cot^2 x + 2\sqrt{3}\cot x + 4\operatorname{cosec} x + 8 = 0$ is : (where $n \in I$)

(a) $n\pi - \frac{\pi}{6}$ (b) $n\pi + \frac{\pi}{6}$ (c) $2n\pi - \frac{\pi}{6}$ (d) $2n\pi + \frac{\pi}{6}$

10. The general solution of the equation $\sin^2 x + \cos^2 3x = 1$ is equal to :

(a) $x = \frac{n\pi}{2}$ (b) $x = n\pi + \frac{\pi}{4}$ (c) $x = \frac{n\pi}{4}$ (d) $x = n\pi + \frac{\pi}{2}$

(where $n \in I$)

11. Values of x between 0 and 2π which satisfy the equation $\sin x\sqrt{8\cos^2 x} = 1$ are in A.P. whose common difference is :

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{2}$ (d) $\frac{2\pi}{3}$

12. Number of solutions of $\sum_{r=1}^{5} \cos rx = 5$ in the interval $[0, 4\pi]$ is :

(a) 0 (b) 2 (c) 3 (d) 7

13. General solution of $4\sin^2 x + \tan^2 x + \operatorname{cosec}^2 x + \cot^2 x - 6 = 0$ is :

(a) $n\pi \pm \frac{\pi}{4}$ (b) $2n\pi \pm \frac{\pi}{4}$ (c) $n\pi + \frac{\pi}{3}$ (d) $n\pi - \frac{\pi}{6}$

[where $n \in I$]

14. Smallest positive x satisfying the equation $\cos^3 3x + \cos^3 5x = 8\cos^3 4x \cdot \cos^3 x$ is :

(a) 15° (b) 18° (c) 22.5° (d) 30°

15. The general solution of the equation $\sin^{100} x - \cos^{100} x = 1$ is (where $n \in I$) :

(a) $2n\pi + \frac{\pi}{2}$ (b) $n\pi + \frac{\pi}{2}$ (c) $2n\pi - \frac{\pi}{2}$ (d) $n\pi$

16. Number of solution(s) of equation $\sin\theta = \sec^2 4\theta$ in $[0, \pi]$ is/are:

(a) 0 (b) 1 (c) 2 (d) 3

17. The number of solutions of the equation $4\sin^2 x + \tan^2 x + \cot^2 x + \operatorname{cosec}^2 x = 6$ in $[0, 2\pi]$

(a) 1 (b) 2 (c) 3 (d) 4

18. The number of solutions of the equation $\sin^4\theta - 2\sin^2\theta - 1 = 0$ which lie between 0 and 2π is :

(a) 0 (b) 2 (c) 4 (d) 8

19. The smallest positive value of p for which the equation $\cos(p\sin x) = \sin(p\cos x)$ has solution in $0 \le x \le 2\pi$ is :

(a) $\frac{\pi}{\sqrt{2}}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{2\sqrt{2}}$ (d) $\frac{3\pi}{2\sqrt{2}}$

20. The total number of ordered pairs (x, y) satisfying $|x|+|y| = 2$ and $\sin\left(\frac{\pi x^2}{3}\right) = 1$ is :

(a) 2 (b) 4 (c) 6 (d) 8

21. The complete set of values of $x, x \in \left(-\frac{\pi}{2}, \pi\right)$ satisfying the inequality $\cos 2x > |\sin x|$ is :

(a) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$ (b) $\left(-\frac{\pi}{2}, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$

(c) $\left(-\frac{\pi}{2}, -\frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \pi\right)$ (d) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, \pi\right)$

22. The total number of solution of the equation $\sin^4 x + \cos^4 x = \sin x \cos x$ in $[0, 2\pi]$ is :

(a) 2 (b) 4 (c) 6 (d) 8

23. Number of solution of the equation $\sin\frac{5x}{2} - \sin\frac{x}{2} = 2$ in the interval $[0, 2\pi]$, is :

(a) 1 (b) 2 (c) 0 (d) Infinite

24. In the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. The equation $\log_{\sin\theta} \cos 2\theta = 2$ has

(a) No solution (b) One solution (c) Two solution (d) Infinite solution

25. If α and β are 2 distinct roots of equation $a\cos\theta + b\sin\theta = C$ then $\cos(\alpha + \beta) =$

(a) $\frac{2ab}{a^2 + b^2}$ (b) $\frac{2ab}{a^2 - b^2}$ (c) $\frac{a^2 + b^2}{a^2 - b^2}$ (d) $\frac{a^2 - b^2}{a^2 + b^2}$

Answers

1.	(c)	**2.**	(a)	**3.**	(d)	**4.**	(a)	**5.**	(b)	**6.**	(c)	**7.**	(d)	**8.**	(c)	**9.**	(c)	**10.**	(c)
11.	(a)	**12.**	(c)	**13.**	(a)	**14.**	(b)	**15.**	( )	**16.**	(b)	**17.**	(d)	**18.**	(a)	**19.**	(c)	**20.**	(b)
21.	(d)	**22.**	(a	**23.**	(c)	**24.**	(b)	**25.**	(d)										

Exercise-2 : One or More than One Answer is/are Correct

1. If $2\cos\theta + 2\sqrt{2} = 3\sec\theta$ where $\theta \in (0, 2\pi)$ then which of the following can be correct ?

(a) $\cos\theta = \frac{1}{\sqrt{2}}$ (b) $\tan\theta = 1$ (c) $\sin\theta = -\frac{1}{\sqrt{2}}$ (d) $\cot\theta = -1$

2. In a triangle ABC if $\tan C < 0$ then :

(a) $\tan A \tan B < 1$ (b) $\tan A \tan B > 1$

(c) $\tan A + \tan B + \tan C < 0$ (d) $\tan A + \tan B + \tan C > 0$

3. The inequality $4\sin 3x + 5 \geq 4\cos 2x + 5\sin x$ is true for $x \in$

(a) $\left[-\pi, \frac{3\pi}{2}\right]$ (b) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (c) $\left[\frac{5\pi}{8}, \frac{13\pi}{8}\right]$ (d) $\left[\frac{23\pi}{14}, \frac{41\pi}{14}\right]$

4. The least difference between the roots of the equation $4\cos x(2 - 3\sin^2 x) + \cos 2x + 1 = 0$ $\forall\, x \in R$ is :

(a) equal to $\frac{\pi}{2}$ (b) $> \frac{\pi}{10}$ (c) $< \frac{\pi}{2}$ (d) $< \frac{\pi}{3}$

5. The equation $\cos x \cos 6x = -1$:

(a) has 50 solutions in $[0, 100\pi]$ (b) has 3 solutions in $[0, 3\pi]$

(c) has even number of solutions in $(3\pi, 13\pi)$ (d) has one solution in $\left[\frac{\pi}{2}, \pi\right]$

6. Identify the correct options :

(a) $\frac{\sin 3\alpha}{\cos 2\alpha} > 0$ for $\alpha \in \left(\frac{3\pi}{8}, \frac{23\pi}{48}\right)$ (b) $\frac{\sin 3\alpha}{\cos 2\alpha} < 0$ for $\alpha \in \left(\frac{13\pi}{48}, \frac{14\pi}{48}\right)$

(c) $\frac{\sin 2\alpha}{\cos\alpha} < 0$ for $\alpha \in \left(-\frac{\pi}{2}, 0\right)$ (d) $\frac{\sin 2\alpha}{\cos\alpha} > 0$ for $\alpha \in \left(\frac{13\pi}{48}, \frac{14\pi}{48}\right)$

7. The equation $\sin^4 x + \cos^4 x + \sin 2x + k = 0$ must have real solutions if :

(a) $k = 0$ (b) $|k| \leq \frac{1}{2}$

(c) $-\frac{3}{2} \leq k \leq \frac{1}{2}$ (d) $-\frac{1}{2} \leq k \leq \frac{3}{2}$

8. Let $f(\theta) = \left(\cos\theta - \cos\frac{\pi}{8}\right)\left(\cos\theta - \cos\frac{3\pi}{8}\right)\left(\cos\theta - \cos\frac{5\pi}{8}\right)\left(\cos\theta - \cos\frac{7\pi}{8}\right)$ then :

(a) maximum value of $f(\theta)\ \forall\, \theta \in R$ is $\frac{1}{4}$

(b) maximum value of $f(\theta)\ \forall\, \theta \in R$ is $\frac{1}{8}$

(c) $f(0) = \frac{1}{8}$

(d) Number of principle solutions of $f(\theta) = 0$ is 8

9. If $\dfrac{\sin^2 2x + 4\sin^4 x - 4\sin^2 x \cdot \cos^2 x}{4 - \sin^2 2x - 4\sin^2 x} = \dfrac{1}{9}$ and $0 < x < \pi$. Then the value of x is :

(a) $\dfrac{\pi}{3}$ (b) $\dfrac{\pi}{6}$ (c) $\dfrac{2\pi}{3}$ (d) $\dfrac{5\pi}{6}$

10. The possible value(s) of 'θ' satisfying the equation

$$\sin^2\theta\tan\theta + \cos^2\theta\cot\theta - \sin 2\theta = 1 + \tan\theta + \cot\theta$$

where $\theta \in [0, \pi]$ is/are :

(a) $\dfrac{\pi}{4}$ (b) π (c) $\dfrac{7\pi}{12}$ (d) None of these

11. If $\sin\theta + \sqrt{3}\cos\theta = 6x - x^2 - 11$, $0 \le \theta \le 4\pi$, $x \in R$ holds for

(a) no values of x and θ

(b) one value of x and two values of θ

(c) two values of x and two values of θ

(d) two pairs of values of (x, θ)

Answers

1.	(a, b, c, d)	**2.**	(a, c)	**3.**	(a, b, c, d)	**4.**	(b, c, d)	**5.**	(a, c, d)	**6.**	(a, b, c, d)
7.	(a, b, c)	**8.**	(b, c, d)	**9.**	(b, d)	**10.**	(c)	**11.**	(b, d)		

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

Consider f, g and h be three real valued function defined on R.
Let $f(x) = \sin 3x + \cos x$, $g(x) = \cos 3x + \sin x$ and $h(x) = f^2(x) + g^2(x)$

1. The length of a longest interval in which the function $y = h(x)$ is increasing, is :

(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$

2. General solution of the equation $h(x) = 4$, is :

(a) $(4n+1)\frac{\pi}{8}$ (b) $(8n+1)\frac{\pi}{8}$ (c) $(2n+1)\frac{\pi}{4}$ (d) $(7n+1)\frac{\pi}{4}$

[where $n \in I$]

3. Number of point(s) where the graphs of the two function, $y = f(x)$ and $y = g(x)$ intersects in $[0, \pi]$, is :

(a) 2 (b) 3 (c) 4 (d) 5

Answers

1.	(b)	**2.**	(a)	**3.**	(c)

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	If $\sin x + \cos x = \frac{1}{5}$; then $\lvert 12\tan x\rvert$ is equal to	(P)	2
(B)	Number of values of θ lying in $(-2\pi, \pi)$ and satisfying $\cot\frac{\theta}{2} = (1+\cot\theta)$ is	(Q)	6
(C)	If $2-\sin^4 x + 8\sin^2 x = \alpha$ has solution, then α can be	(R)	9
(D)	Number of integral values of x satisfying $\log_4(2x^2+5x+27)-\log_2(2x-1)\geq 0$	(S)	14
		(T)	16

2.

	Column-I		Column-II
(A)	If $x, y \in [0, 2\pi]$, then total number of ordered pair (x, y) satisfying $\sin x \cos y = 1$ is	(P)	4
(B)	If $f(x) = \sin x - \cos x - kx + b$ decreases for all real values of x, then $2\sqrt{2}\,k$ may be	(Q)	0
(C)	The number of solution of the equation $\sin^{-1}(\lvert x^2-1\rvert)+\cos^{-1}(\lvert 2x^2-5\rvert)=\frac{\pi}{2}$ is	(R)	2
(D)	The number of ordered pair (x, y) satisfying the equation $\sin x + \sin y = \sin(x+y)$ and $\lvert x\rvert+\lvert y\rvert=1$ is	(S)	3
		(T)	6

3.

	Column-I		Column-II
(A)	Minimum value of $y = 4\sec^2 x + \cos^2 x$ for permissible real values of x is equal to	(P)	2
(B)	If m, n are positive integers and $m + n\sqrt{2} = \sqrt{41+24\sqrt{2}}$ then $(m+n)$ is equal to :	(Q)	7

(C)	Number of solutions of the equation : $\log_{\left(\frac{9x-x^2-14}{7}\right)}(\sin 3x - \sin x) = \log_{\left(\frac{9x-x^2-14}{7}\right)} \cos 2x$ is equal to :	(R)	4
(D)	Consider an arithmetic sequence of positive integers. If the sum of the first ten terms is equal to the 58th term, then the least possible value of the first term is equal to :	(S)	5
		(T)	3

Answers

1.	A → R, T ; B → P ; C → P, Q, R ; D → Q
2.	A → S ; B → P, T ; C → R ; D → T
3.	A → S ; B → Q ; C → P ; D → R

Exercise-5 : Subjective Type Problems

1. Find the number of solutins of the equations
   $(\sin x-1)^3+(\cos x-1)^3+(\sin x)^3=(2\sin x+\cos x-2)^3$ in $[0, 2\pi]$.
2. If $x+\sin y=2014$ and $x+2014\cos y=2013$, $0 \le y \le \frac{\pi}{2}$, then find the value of $[x+y]-2005$ (where $[\cdot]$ denotes greatest integer function)
3. The complete set of values of x satisfying $\frac{2\sin 6x}{\sin x-1}<0$ and $\sec^2 x-2\sqrt{2}\tan x \le 0$ in $\left(0, \frac{\pi}{2}\right)$ is $[a, b) \cup (c, d]$, then find the value of $\left(\frac{cd}{ab}\right)$.
4. The range of value's of k for which the equation $2\cos^4 x-\sin^4 x+k=0$ has atleast one solution is $[\lambda, \mu]$. Find the value of $(9\mu+\lambda)$.
5. The number of points in interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, where the graphs of the curves $y=\cos x$ and $y=\sin 3x$, $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ intersects is
6. The number of solutions of the system of equations :
   $$2\sin^2 x+\sin^2 2x=2$$
   $$\sin 2x+\cos 2x=\tan x$$
   in $[0, 4\pi]$ satisfying $2\cos^2 x+\sin x \le 2$ is :
7. If the sum of all the solutions of the equation $3\cot^2\theta+10\cot\theta+3=0$ in $[0, 2\pi]$ is $k\pi$ where $k \in I$, then find the value of k.
8. If the sum of all values of θ, $0 \le \theta \le 2\pi$ satisfying the equation
   $(8\cos 4\theta-3)(\cot\theta+\tan\theta-2)(\cot\theta+\tan\theta+2)=12$ is $k\pi$, then k is equal to :
9. Find the number of solutions of the equation $2\sin^2 x+\sin^2 2x=2$; $\sin 2x+\cos 2x=\tan x$ in $[0, 4\pi]$ satisfying the condition $2\cos^2 x+\sin x \le 2$.

Answers

1.	5	**2.**	9	**3.**	6	**4.**	7	**5.**	3	**6.**	8	**7.**	5
8.	8	**9.**	8										

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Chapter 24 – Solution of Triangles

24 Solution of Triangles

Exercise-1 : Single Choice Problems

1. In a ΔABC if $9(a^2+b^2)=17c^2$ then the value of the expression $\frac{\cot A+\cot B}{\cot C}$ is :

(a) $\frac{13}{4}$ (b) $\frac{7}{4}$ (c) $\frac{5}{4}$ (d) $\frac{9}{4}$

2. Let H be the orthocenter of triangle ABC, then angle subtended by side BC at the centre of incircle of ΔCHB is :

(a) $\frac{A}{2}+\frac{\pi}{2}$ (b) $\frac{B+C}{2}+\frac{\pi}{2}$ (c) $\frac{B-C}{2}+\frac{\pi}{2}$ (d) $\frac{B+C}{2}+\frac{\pi}{4}$

3. Circum radius of a ΔABC is 3 units; let O be the circum centre and H be the orthocentre then the value of $\frac{1}{64}(AH^2+BC^2)(BH^2+AC^2)(CH^2+AB^2)$ equals :

(a) 3^4 (b) 9^3 (c) 27^6 (d) 81^4

4. The angles A, B and C of a triangle ABC are in arithmetic progression. If $2b^2=3c^2$ then the angle A is :

(a) $15°$ (b) $60°$ (c) $75°$ (d) $90°$

5. In a triangle ABC, if $\tan\frac{A}{2}\tan\frac{C}{2}=\frac{1}{3}$ and $ac=4$, then the least value of b is :

(notation have their usual meaning)

(a) 1 (b) 2 (c) 4 (d) 6

6. In a triangle ABC the expression $a\cos B\cos C+b\cos C\cos A+c\cos A\cos B$ equals to :

(a) $\frac{rs}{R}$ (b) $\frac{r}{sR}$ (c) $\frac{R}{rs}$ (d) $\frac{Rs}{r}$

7. The set of real numbers a such that $a^2+2a, 2a+3, a^2+3a+8$ are the sides of a triangle, is :

(a) $(0,\infty)$ (b) $(5,8)$ (c) $\left(-\frac{11}{3},\infty\right)$ (d) $(5,\infty)$

8. In a ΔABC, $\angle B = \frac{\pi}{3}$ and $\angle C = \frac{\pi}{4}$ let D divide BC internally in the ratio $1:3$, then $\frac{\sin(\angle BAD)}{\sin(\angle CAD)}$ is equal to :

(a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\frac{\sqrt{2}}{3}$

9. Let AD, BE, CF be the lengths of internal bisectors of angles A, B, C respectively of triangle ABC. Then the harmonic mean of $AD\sec\frac{A}{2}, BE\sec\frac{B}{2}, CF\sec\frac{C}{2}$ is equal to :

(a) Harmonic mean of sides of ΔABC
(b) Geometric mean of sides of ΔABC
(c) Arithmetic mean of sides of ΔABC
(d) Sum of reciprocals of the sides of ΔABC

10. In triangle ABC, if $2b = a + c$ and $A - C = 90°$, then $\sin B$ equals :
[**Note :** All symbols used have usual meaning in triangle ABC.]

(a) $\frac{\sqrt{7}}{5}$ (b) $\frac{\sqrt{5}}{8}$ (c) $\frac{\sqrt{7}}{4}$ (d) $\frac{\sqrt{5}}{3}$

11. In a triangle ABC, if $2a\cos\left(\frac{B-C}{2}\right) = b + c$, then $\sec A$ is equal to :

(All symbols used have usual meaning in a triangle.)

(a) $\frac{2}{\sqrt{3}}$ (b) $\sqrt{2}$ (c) 2 (d) 3

12. Triangle ABC has $BC = 1$ and $AC = 2$, then maximum possible value of $\angle A$ is :

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

13. $\Delta I_1I_2I_3$ is an excentral triangle of an equilateral triangle ΔABC such that $I_1I_2 = 4$ unit, if ΔDEF is pedal triangle of ΔABC, then $\frac{Ar(\Delta I_1I_2I_3)}{Ar(\Delta DEF)} =$

(a) 16 (b) 4 (c) 2 (d) 1

14. Let ABC be a triangle with $\angle BAC = \frac{2\pi}{3}$ and $AB = x$ such that $(AB)(AC) = 1$. If x varies then the longest possible length of the internal angle bisector AD equals :

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{\sqrt{2}}{3}$

15. In an equilateral triangle r, R and r_1 form (where symbols used have usual meaning)

(a) an A.P. (b) a G.P. (c) an H.P. (d) none of these

16. In ΔABC if $\frac{\sin A}{\sin C} = \frac{\sin(A-B)}{\sin(B-C)}$, then a^2, b^2, c^2 are in :

(a) A.P. (b) G.P. (c) H.P. (d) none of these

17. In ΔABC, $\tan A = 2$, $\tan B = \frac{3}{2}$ and $c = \sqrt{65}$, then circumradius of the triangle is :

(a) 65 (b) $\frac{65}{7}$ (c) $\frac{65}{14}$ (d) none of these

18. If the sides a, b, c of a triangle ABC are the roots of the equation $x^3 - 13x^2 + 54x - 72 = 0$, then the value of $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c}$ is equal to :

(a) $\frac{61}{144}$ (b) $\frac{61}{72}$ (c) $\frac{169}{144}$ (d) $\frac{59}{144}$

19. In ΔABC, if $\angle C = 90°$, then $\frac{a+c}{b} + \frac{b+c}{a}$ is equal to :

(a) $\frac{c}{r}$ (b) $\frac{1}{2Rr}$ (c) 2 (d) $\frac{R}{r}$

20. In a ΔABC, if $a^2 \sin B = b^2 + c^2$, then :

(a) $\angle A$ is obtuse (b) $\angle A$ is acute (c) $\angle B$ is obtuse (d) $\angle A$ is right angle

21. If R and R' are the circumradii of triangles ABC and OBC, where O is the orthocenter of triangle ABC, then :

(a) $R' = \frac{R}{2}$ (b) $R' = 2R$ (c) $R' = R$ (d) $R' = 3R$

22. The acute angle of a rhombus whose side is geometric mean between its diagonals, is :

(a) 15° (b) 20° (c) 30° (d) 60°

23. In a ΔABC right angled at A, a line is drawn through A to meet BC at D dividing BC in $2 : 1$. If $\tan(\angle ADC) = 3$ then $\angle BAD$ is :

(a) 30° (b) 45° (c) 60° (d) 75°

24. A circle is cirumscribed in an equilateral triangle of side 'l'. The area of any square inscribed in the circle is :

(a) $\frac{4}{3}l^2$ (b) $\frac{2}{3}l^2$ (c) $\frac{1}{3}l^2$ (d) l^2

25. If the sides of a triangle are in the ratio $2 : \sqrt{6} : (\sqrt{3}+1)$, then the largest angle of the triangle will be :

(a) 60° (b) 72° (c) 75° (d) 90°

26. In a triangle ABC if a, b, c are in A.P. and $C - A = 120°$, then $\frac{s}{r} =$

(where notations have their usual meaning)

(a) $\sqrt{15}$ (b) $2\sqrt{15}$ (c) $3\sqrt{15}$ (d) $6\sqrt{15}$

27. In a triangle ABC, $a = 5, b = 4$ and $\cos(A - B) = \frac{31}{32}$, then the third side is equal to :

(where symbols used have usual meanings)

(a) $\sqrt{6}$ (b) $6\sqrt{6}$ (c) 6 (d) $(216)^{1/4}$

28. If semiperimeter of a triangle is 15, then the value of $(b+c)\cos(B+C)+(c+a)\cos(C+A)+(a+b)\cos(A+B)$ is equal to :
(where symbols used have usual meanings)
(a) -60 (b) -15
(c) -30 (d) can not be determined

29. Let triangle ABC be an isosceles triangle with $AB = AC$. Suppose that the angle bisector of its angle B meets the side AC at a point D and that $BC = BD + AD$. Measure of the angle A in degrees, is :
(a) 80 (b) 100 (c) 110 (d) 130

30. In triangle ABC if $A:B:C = 1:2:4$, then $(a^2-b^2)(b^2-c^2)(c^2-a^2) = \lambda a^2b^2c^2$, where $\lambda =$
(where notations have their usual meaning)
(a) 1 (b) 2 (c) 4 (d) 9

31. In a triangle ABC with altitude AD, $\angle BAC = 45°$, $DB = 3$ and $CD = 2$. The area of the triangle ABC is :
(a) 6 (b) 15 (c) 15/4 (d) 12

32. A triangle has base 10 cm long and the base angles of 50° and 70°. If the perimeter of the triangle is $x + y\cos z°$ where $z \in (0, 90)$ then the value of $x+y+z$ equals :
(a) 60 (b) 55 (c) 50 (d) 40

33. Let H be the orthocenter of triangle ABC, then angle subtended by side BC at the centre of incircle of ΔCHB is :
(a) $\frac{A}{2}+\frac{\pi}{2}$ (b) $\frac{B+C}{2}+\frac{\pi}{2}$ (c) $\frac{B-C}{2}+\frac{\pi}{2}$ (d) $\frac{B+C}{2}+\frac{\pi}{4}$

34. Triangle ABC is right angled at A. The points P and Q are on the hypotenuse BC such that $BP = PQ = QC$. If $AP = 3$ and $AQ = 4$ then the length BC is equal to :
(a) $\sqrt{27}$ (b) $\sqrt{36}$ (c) $\sqrt{45}$ (d) $\sqrt{54}$

35. In a ΔABC if $b = a(\sqrt{3}-1)$ and $\angle C = 30°$ then the measure of the angle A is :
(a) 15° (b) 45° (c) 75° (d) 105°

36. Through the centroid of an equilateral triangle, a line parallel to the base is drawn. On this line, an arbitrary point P is taken inside the triangle. Let h denote the perpendicular distance of P from the base of the triangle. Let h_1 and h_2 be the perpendicular distance of P from the other two sides of the triangle. Then :
(a) $h = \frac{h_1+h_2}{2}$ (b) $h = \sqrt{h_1h_2}$
(c) $h = \frac{2h_1h_2}{h_1+h_2}$ (d) $h = \frac{(h_1+h_2)\sqrt{3}}{4}$

37. The angles A, B and C of a triangle ABC are in arithmetic progression. $AB = 6$ and $BC = 7$. Then AC is :
(a) $\sqrt{41}$ (b) $\sqrt{39}$ (c) $\sqrt{42}$ (d) $\sqrt{43}$

38. In ΔABC, If $A - B = 120°$ and $R = 8r$, then the value of $\frac{1+\cos C}{1-\cos C}$ equals :

(All symbols used have their usual meaning in a triangle)

(a) 12 (b) 15 (c) 21 (d) 31

39. The lengths of the sides CB and CA of a triangle ABC are given by a and b and the angle C is $\frac{2\pi}{3}$.

The line CD bisects the angle C and meets AB at D. Then the length of CD is :

(a) $\frac{1}{a+b}$ (b) $\frac{a^2+b^2}{a+b}$ (c) $\frac{ab}{2(a+b)}$ (d) $\frac{ab}{a+b}$

40. In ΔABC, angle A is $120°$, $BC + CA = 20$ and $AB + BC = 21$, then the length of the side BC, equals :

(a) 13 (b) 15 (c) 17 (d) 19

41. A triangle has sides 6, 7, 8. The line through its incentre parallel to the shortest side is drawn to meet the other two sides at P and Q. The length of the segment PQ is :

(a) $\frac{12}{5}$ (b) $\frac{15}{4}$ (c) $\frac{30}{7}$ (d) $\frac{33}{9}$

42. The perimeter of a ΔABC is 48 cm and one side is 20 cm. Then remaining sides of ΔABC must be greater than :

(a) 8 cm (b) 9 cm (c) 12 cm (d) 4 cm

43. In an equilateral ΔABC, (where symbols used have usual meanings), then r, R and r_1 form :

(a) an A.P. (b) a G.P.

(c) an H.P. (d) neither an A.P., G.P. nor H.P.

44. The expression $\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$ is equal to :

(a) $\cos^2 A$ (b) $\sin^2 A$ (c) $\cos A \cos B \cos C$ (d) $\sin A \sin B \sin C$

(where symbols used have usual meanings)

45. Circumradius of an isosceles ΔABC with $\angle A = \angle B$ is 4 times its in radius, then $\cos A$ is root of the equation :

(a) $x^2 - x - 8 = 0$ (b) $8x^2 - 8x + 1 = 0$ (c) $x^2 - x - 4 = 0$ (d) $4x^2 - 4x + 1 = 0$

46. A is the orthocentre of ΔABC and D is reflection point of A w.r.t. perpendicular bisector of BC, then orthocenter of ΔDBC is :

(a) D (b) C (c) B (d) A

47. If a, b, c are sides of a scalene triangle, then the value of determinant $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is always:

(a) ≥ 0 (b) > 0 (c) ≤ -1 (d) < 0

48. In a triangle ABC if $A : B : C = 1 : 2 : 4$, then $(a^2 - b^2)(b^2 - c^2)(c^2 - a^2) = \lambda\, a^2b^2c^2$, where $\lambda =$:

(a) 1 (b) 2 (c) 3 (d) $\frac{1}{3}$

49. The minimum value of $\frac{r_1 r_2 r_3}{r^3}$ in a triangle is (symbols have their usual meaning)

(a) 1 (b) 3 (c) 8 (d) 27

50. In a triangle ABC, $BC = 3$, $AC = 4$ and $AB = 5$. The value of $\sin A + \sin 2B + \sin 3C$ equals

(a) $\frac{24}{25}$ (b) $\frac{14}{25}$ (c) $\frac{64}{25}$ (d) None

51. In any triangle ABC, the value of $\frac{r_1 + r_2}{1 + \cos C}$ is equal to (where notation have their usual meaning) :

(a) $2R$ (b) $2r$ (c) R (d) $\frac{2R^2}{r}$

52. In a triangle ABC, medians AD and BE are drawn. If $AD = 4$; $\angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$ then the area of the triangle ABC is :

(a) $\frac{8}{3\sqrt{3}}$ (b) $\frac{16}{3\sqrt{3}}$ (c) $\frac{32}{3\sqrt{3}}$ (d) $\frac{64}{3\sqrt{3}}$

53. The sides of a triangle are $\sin\alpha$, $\cos\alpha$, $\sqrt{1 + \sin\alpha\cos\alpha}$ for some $0 < \alpha < \frac{\pi}{2}$ then the greatest angle of the triangle is :

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

54. Let ABC be a right triangle with $\angle BAC = \frac{\pi}{2}$, then $\left(\frac{r^2}{2R^2} + \frac{r}{R}\right)$ is equal to :

(where symbols used have usual meaning in a triangle)

(a) $\sin B \sin C$ (b) $\tan B \tan C$ (c) $\sec B \sec C$ (d) $\cot B \cot C$

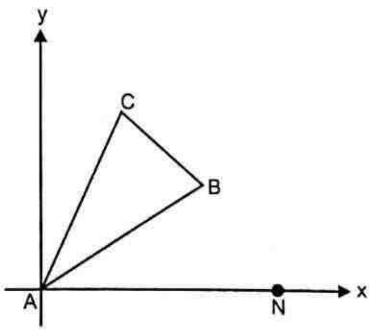
55. Find the radius of the circle escribed to the triangle ABC (Shown in the figure below) on the side BC if $\angle NAB = 30°$; $\angle BAC = 30°$; $AB = AC = 5$.

(a) $\frac{(10\sqrt{2} + 5\sqrt{3} - 5)(2 - \sqrt{3})}{2\sqrt{2}}$

(b) $\frac{(10\sqrt{2} + 5\sqrt{3} + 5)}{2\sqrt{2}}(2 - \sqrt{3})$

(c) $\frac{(10\sqrt{2} + 5\sqrt{3} - 5)}{2\sqrt{2}}(2 + \sqrt{3})$

(d) $\frac{(10\sqrt{2} + 5\sqrt{2} + 1)}{2\sqrt{3}}(\sqrt{3} - 1)$



56. In a ΔABC, with usual notations, if $b > c$ then distance between foot of median and foot of altitude both drawn from vertex A on BC is :

(a) $\frac{a^2 - b^2}{2c}$ (b) $\frac{b^2 - c^2}{2a}$ (c) $\frac{b^2 + c^2 - a^2}{2a}$ (d) $\frac{b^2 + c^2 - a^2}{2c}$

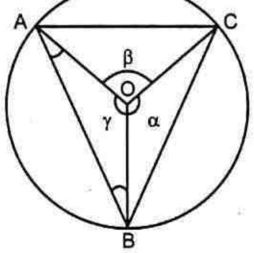
57. In a triangle ABC the expression $a\cos B\cos C + b\cos C\cos A + c\cos A\cos B$ equals to :

(a) $\frac{rs}{R}$ (b) $\frac{r}{sR}$ (c) $\frac{R}{rs}$ (d) $\frac{Rs}{r}$

58. In an acute triangle ABC, altitudes from the vertices A, B and C meet the opposite sides at the points D, E and F respectively. If the radius of the circumcircle of ΔAFE, ΔBFD, ΔCED, ΔABC be respectively R_1, R_2, R_3 and R. Then the maximum value of $R_1 + R_2 + R_3$ is :

(a) $\frac{3R}{8}$ (b) $\frac{2R}{3}$ (c) $\frac{4R}{3}$ (d) $\frac{3R}{2}$

59. A circle of area 20 sq. units is centered at the point O. Suppose ΔABC is inscribed in that circle and has area 8 sq. units. The central angles α, β and γ are as shown in the figure. The value of $(\sin\alpha + \sin\beta + \sin\gamma)$ is equal to :



(a) $\frac{4\pi}{5}$ (b) $\frac{3\pi}{4}$ (c) $\frac{2\pi}{5}$ (d) $\frac{\pi}{4}$

Answers

1.	(d)	**2.**	(b)	**3.**	(b)	**4.**	(c)	**5.**	(b)	**6.**	(a)	**7.**	(d)	**8.**	(a)	**9.**	(a)	**10.**	(c)
11.	(c)	**12.**	(a)	**13.**	(a)	**14.**	(b)	**15.**	(a)	**16.**	(a)	**17.**	(c)	**18.**	(a)	**19.**	(a)	**20.**	(a)
21.	(c)	**22.**	(c)	**23.**	(b)	**24.**	(b)	**25.**	(c)	**26.**	(c)	**27.**	(c)	**28.**	(c)	**29.**	(b)	**30.**	(a)
31.	(b)	**32.**	(d)	**33.**	(b)	**34.**	(c)	**35.**	(d)	**36.**	(a)	**37.**	(d)	**38.**	(b)	**39.**	(d)	**40.**	(a)
41.	(c)	**42.**	(d)	**43.**	(a)	**44.**	(b)	**45.**	(b)	**46.**	(a)	**47.**	(d)	**48.**	(a)	**49.**	(d)	**50.**	(b)
51.	(a)	**52.**	(c)	**53.**	(c)	**54.**	(a)	**55.**	(a)	**56.**	(b)	**57.**	(a)	**58.**	(d)	**59.**	(a)		

Exercise-2 : One or More than One Answer is/are Correct

1. If r_1, r_2, r_3 are radii of the escribed circles of a triangle ABC and r is the radius of its incircle, then the root(s) of the equation $x^2 - r(r_1r_2 + r_2r_3 + r_3r_1)x + (r_1r_2r_3 - 1) = 0$ is/are :

(a) r_1 (b) $r_2 + r_3$ (c) 1 (d) $r_1r_2r_3 - 1$

2. In ΔABC, $\angle A = 60°$, $\angle B = 90°$, $\angle C = 30°$. Let H be its orthocentre, then :
(where symbols used have usual meanings)

(a) $AH = c$ (b) $CH = a$ (c) $AH = a$ (d) $BH = 0$

3. In an equilateral triangle, if inradius is a rational number then which of the following is/are correct ?

(a) circumradius is always rational (b) exradii are always rational
(c) area is always ir-rational (d) perimeter is always rational

4. Let A, B, C be angles of a triangle ABC and let $D = \frac{5\pi + A}{32}, E = \frac{5\pi + B}{32}, F = \frac{5\pi + C}{32}$, then :

$\left(\text{where } D, E, F \neq \frac{n\pi}{2}, n \in I, I \text{ denote set of integers}\right)$

(a) $\cot D \cot E + \cot E \cot F + \cot D \cot F = 1$ (b) $\cot D + \cot E + \cot F = \cot D \cot E \cot F$
(c) $\tan D \tan E + \tan E \tan F + \tan F \tan D = 1$ (d) $\tan D + \tan E + \tan F = \tan D \tan E \tan F$

5. In a triangle ABC, if $a = 4, b = 8$ and $\angle C = 60°$, then :
(where symbols used have usual meanings)

(a) $c = 6$ (b) $c = 4\sqrt{3}$ (c) $\angle A = 30°$ (d) $\angle B = 90°$

6. In a ΔABC if $\frac{r}{r_1} = \frac{r_2}{r_3}$, then which of the following is/are true ?
(where symbols used have usual meanings)

(a) $a^2 + b^2 + c^2 = 8R^2$ (b) $\sin^2 A + \sin^2 B + \sin^2 C = 2$
(c) $a^2 + b^2 = c^2$ (d) $\Delta = s(s + c)$

7. ABC is a triangle whose circumcentre, incentre and orthocentre are O, I and H respectively which lie inside the triangle, then :

(a) $\angle BOC = A$ (b) $\angle BIC = \frac{\pi}{2} + \frac{A}{2}$
(c) $\angle BHC = \pi - A$ (d) $\angle BHC = \pi - \frac{A}{2}$

8. In a triangle ABC, $\tan A$ and $\tan B$ satisfy the inequality $\sqrt{3}x^2 - 4x + \sqrt{3} < 0$, then which of the following must be correct ?
(where symbols used have usual meanings)

(a) $a^2 + b^2 - ab < c^2$ (b) $a^2 + b^2 > c^2$
(c) $a^2 + b^2 + ab > c^2$ (d) $a^2 + b^2 < c^2$

9. If in a ΔABC ; $\angle C = \frac{\pi}{8}$; $a = \sqrt{2}$; $b = \sqrt{2+\sqrt{2}}$ then the measure of $\angle A$ can be :

(a) 45° (b) 135° (c) 30° (d) 150°

10. In triangle ABC, $a = 3, b = 4, c = 2$. Point D and E trisect the side BC. If $\angle DAE = \theta$, then $\cot^2\theta$ is divisible by :

(a) 2 (b) 3 (c) 5 (d) 7

11. In a ΔABC if $3\sin A + 4\cos B = 6$; $4\sin B + 3\cos A = 1$ then possible value(s) of $\angle C$ be:

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{3}$ (d) $\frac{5\pi}{6}$

12. If the line joining the incentre to the centroid of a triangle ABC is parallel to the side BC. Which of the following are correct ?

(a) $2b = a + c$ (b) $2a = b + c$ (c) $\cot\frac{A}{2}\cot\frac{C}{2} = 3$ (d) $\cot\frac{B}{2}\cot\frac{C}{2} = 3$

13. In a triangle the length of two larger sides are 10 and 9 respectively. It the angles are in A.P., the length of third side can be :

(a) $5-\sqrt{6}$ (b) $5+\sqrt{6}$ (c) $6-\sqrt{5}$ (d) $6+\sqrt{5}$

14. If area of ΔABC, Δ and angle C are given and if the side c opposite to given angle is minimum, then

(a) $a = \sqrt{\frac{2\Delta}{\sin C}}$ (b) $b = \sqrt{\frac{2\Delta}{\sin C}}$ (c) $a = \frac{4\Delta}{\sin C}$ (d) $b = \frac{4\Delta}{\sin^2 C}$

15. In a triangle ABC, if $\tan A = 2\sin 2C$ and $3\cos A = 2\sin B\sin C$ then possible values of C is/are

(a) $\frac{\pi}{8}$ (b) $\frac{\pi}{6}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

Answers

1.	(c, d)	**2.**	(a, b, d)	**3.**	(a, b, c)	**4.**	(b, c)	**5.**	(b, c, d)	**6.**	(a, b, c)
7.	(b, c)	**8.**	(a, c)	**9.**	(a)	**10.**	(b, c)	**11.**	(b)	**12.**	(b, d)
13.	(a, b)	**14.**	(a, b)	**15.**	(c, d)						

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $\angle A = 23°$, $\angle B = 75°$ and $\angle C = 82°$ be the angles of ΔABC.
The incircle of ΔABC touches the sides BC, CA, AB at points D, E, F respectively. Let r', r_1' respectively be the inradius, exradius opposite to vertex D of ΔDEF and r be the inradius of ΔABC, then

1. $\dfrac{r'}{r} =$

(a) $\sin\dfrac{A}{2} + \sin\dfrac{B}{2} + \sin\dfrac{C}{2} - 1$
(b) $1 - \sin\dfrac{A}{2} + \sin\dfrac{B}{2} + \sin\dfrac{C}{2}$
(c) $\cos\dfrac{A}{2} + \cos\dfrac{B}{2} + \cos\dfrac{C}{2} - 1$
(d) $1 - \cos\dfrac{A}{2} + \cos\dfrac{B}{2} + \cos\dfrac{C}{2}$

2. $\dfrac{r_1'}{r} =$

(a) $\sin\dfrac{A}{2} + \sin\dfrac{B}{2} + \sin\dfrac{C}{2} - 1$
(b) $1 - \sin\dfrac{A}{2} + \sin\dfrac{B}{2} + \sin\dfrac{C}{2}$
(c) $\cos\dfrac{A}{2} + \cos\dfrac{B}{2} + \cos\dfrac{C}{2} - 1$
(d) $1 - \cos\dfrac{A}{2} + \cos\dfrac{B}{2} + \cos\dfrac{C}{2}$

Paragraph for Question Nos. 3 to 4

Internal angle bisectors of ΔABC meets its circum circle at D, E and F where symbols have usual meaning.

3. Area of ΔDEF is :

(a) $2R^2 \cos^2\left(\dfrac{A}{2}\right)\cos^2\left(\dfrac{B}{2}\right)\cos^2\left(\dfrac{C}{2}\right)$
(b) $2R^2 \sin\left(\dfrac{A}{2}\right)\sin\left(\dfrac{B}{2}\right)\sin\left(\dfrac{C}{2}\right)$
(c) $2R^2 \sin^2\left(\dfrac{A}{2}\right)\sin^2\left(\dfrac{B}{2}\right)\sin^2\left(\dfrac{C}{2}\right)$
(d) $2R^2 \cos\left(\dfrac{A}{2}\right)\cos\left(\dfrac{B}{2}\right)\cos\left(\dfrac{C}{2}\right)$

4. The ratio of area of triangle ABC and triangle DEF is :

(a) ≥ 1 (b) ≤ 1 (c) $\geq 1/2$ (d) $\leq 1/2$

Paragraph for Question Nos. 5 to 6

Let triangle ABC is right triangle right angled at C such that $A < B$ and $r = 8$, $R = 41$.

5. Area of ΔABC is :

(a) 720 (b) 1440 (c) 360 (d) 480

6. $\tan\dfrac{A}{2} =$

(a) $\dfrac{1}{18}$ (b) $\dfrac{1}{3}$ (c) $\dfrac{1}{6}$ (d) $\dfrac{1}{9}$

[where notations have their usual meaning]

Paragraph for Question Nos. 7 to 8

Let the incircle of ΔABC touches the sides BC, CA, AB at A_1, B_1, C_1 respectively. The incircle of $\Delta A_1B_1C_1$ touches its sides of B_1C_1, C_1A_1 and A_1B_1 at A_2, B_2, C_2 respectively and so on.

7. $\lim\limits_{n\to\infty} \angle A_n =$

(a) 0 (b) $\dfrac{\pi}{6}$ (c) $\dfrac{\pi}{4}$ (d) $\dfrac{\pi}{3}$

8. In $\Delta A_4B_4C_4$, the value of $\angle A_4$ is :

(a) $\dfrac{3\pi+A}{6}$ (b) $\dfrac{3\pi-A}{8}$ (c) $\dfrac{5\pi-A}{16}$ (d) $\dfrac{5\pi+A}{16}$

Paragraph for Question Nos. 9 to 10

Let ABC be a given triangle. Points D and E are on sides AB and AC respectively and point F is on line segment DE. Let $\dfrac{AD}{AB} = x, \dfrac{AE}{AC} = y, \dfrac{DF}{DE} = z$. Let area of $\Delta BDF = \Delta_1$, area of $\Delta CEF = \Delta_2$ and area of $\Delta ABC = \Delta$.

9. $\dfrac{\Delta_1}{\Delta}$ is equal to :

(a) xyz (b) $(1-x)y(1-z)$ (c) $(1-x)yz$ (d) $x(1-y)z$

10. $\dfrac{\Delta_2}{\Delta}$ is equal to :

(a) $(1-x)y(1-z)$ (b) $(1-x)(1-y)z$ (c) $x(1-y)(1-z)$ (d) $(1-x)yz$

Paragraph for Question Nos. 11 to 13

a, b, c are the length of sides BC, CA, AB respectively of ΔABC satisfying $\log\left(1+\dfrac{c}{a}\right) + \log a - \log b = \log 2$.

Also the quadratic equation $a(1-x^2) + 2bx + c(1+x^2) = 0$ has two equal roots.

11. a, b, c are in :

(a) A.P. (b) G.P. (c) H.P. (d) None

12. Measure of angle C is :

(a) 30° (b) 45° (c) 60° (d) 90°

13. The value of $(\sin A + \sin B + \sin C)$ is equal to :

(a) $\frac{5}{2}$ (b) $\frac{12}{5}$

(c) $\frac{8}{3}$ (d) 2

Paragraph for Question Nos. 14 to 16

Let ABC be a triangle inscribed in a circle and let $l_a = \frac{m_a}{M_a}$; $l_b = \frac{m_b}{M_b}$; $l_c = \frac{m_c}{M_c}$ where m_a, m_b, m_c are the lengths of the angle bisectors of angles A, B and C respectively, internal to the triangle and M_a, M_b and M_c are the lengths of these internal angle bisectors extended until they meet the circumcircle.

14. l_a equals :

(a) $\dfrac{\sin A}{\sin\left(B + \frac{A}{2}\right)}$ (b) $\dfrac{\sin B \sin C}{\sin^2\left(\frac{B+C}{2}\right)}$ (c) $\dfrac{\sin B \sin C}{\sin^2\left(B + \frac{A}{2}\right)}$ (d) $\dfrac{\sin B + \sin C}{\sin^2\left(B + \frac{A}{2}\right)}$

15. The maximum value of the product $(l_a l_b l_c) \times \cos^2\left(\frac{B-C}{2}\right) \times \cos^2\left(\frac{C-A}{2}\right) \times \cos^2\left(\frac{A-B}{2}\right)$ is equal to :

(a) $\frac{1}{8}$ (b) $\frac{1}{64}$ (c) $\frac{27}{64}$ (d) $\frac{27}{32}$

16. The minimum value of the expression $\dfrac{l_a}{\sin^2 A} + \dfrac{l_b}{\sin^2 B} + \dfrac{l_c}{\sin^2 C}$ is :

(a) 2 (b) 3 (c) 4 (d) none of these

Answers

1.	(a)	**2.**	(b)	**3.**	(d)	**4.**	(b)	**5.**	(a)	**6.**	(d)	**7.**	(d)	**8.**	(d)	**9.**	(c)	**10.**	(c)
11.	(a)	**12.**	(d)	**13.**	(b)	**14.**	(c)	**15.**	(c)	**16.**	(b)								

Exercise-4 : Matching Type Problems

1. Consider a right angled triangle ***ABC* right angled at *C*** with integer sides. List-I gives inradius. List-II gives the number of triangles.

	Column-I		Column-II
(A)	3	**(P)**	6
(B)	4	**(Q)**	7
(C)	6	**(R)**	8
(D)	9	**(S)**	10
		(T)	12

2.

	Column-I		Column-II
(A)	Find the sum of the series $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{6}+\frac{1}{8}+\frac{1}{9}+\frac{1}{12}+\ldots\ldots\infty$, where the terms are the reciprocals of the positive integers whose only prime factors are two's and three's	**(P)**	7
(B)	The length of the sides of ΔABC are a, b and c and A is the angle opposite to side a. If $b^2+c^2=a^2+54$ and $bc=\frac{a^3}{\cos A}$ then the value of $\left(\frac{b^2+c^2}{9}\right)$, is	**(Q)**	10
(C)	The equations of perpendicular bisectors of two sides AB and AC of a triangle ABC are $x+y+1=0$ and $x-y+1=0$ respectively. If circumradius of ΔABC is 2 units and the locus of vertex A is $x^2+y^2+gx+c=0$, then (g^2+c^2), is equal to	**(R)**	13
(D)	Number of solutions of the equation $\cos\theta\sin\theta+6(\cos\theta-\sin\theta)+6=0$ in $[0, 30]$, is equal to	**(S)**	3

3. In ΔABC, if $r_1=21, r_2=24, r_3=28$, then

	Column-I		Column-II
(A)	$a=$	**(P)**	8
(B)	$b=$	**(Q)**	12
(C)	$s=$	**(R)**	26

(D)	$r =$	**(S)**	28
		(T)	42

(Where notations have their usual meaning)

4.

	Column-I		Column-II
(A)	$\dfrac{r_1(r_2+r_3)}{\sqrt{r_2r_3+r_3r_1+r_1r_2}}$	**(P)**	$\sin\dfrac{A}{2}$
(B)	$\dfrac{r_1}{\sqrt{(r_1+r_2)(r_1+r_3)}}$	**(Q)**	$4R$
(C)	$r_1+r_2+r_3-r$	**(R)**	0
(D)	$\dfrac{1}{r_1}+\dfrac{1}{r_2}+\dfrac{1}{r_3}-\dfrac{1}{r}$	**(S)**	$2R\sin A$

Answers

1.	$A\to P$; $B\to P$; $C\to T$; $D\to S$
2.	$A\to S$; $B\to P$; $C\to R$; $D\to Q$
3.	$A\to R$; $B\to S$; $C\to T$; $D\to P$
4.	$A\to S$; $B\to P$; $C\to Q$; $D\to R$

Exercise-5 : Subjective Type Problems

1. If the median AD of ΔABC makes an angle $\angle ADC = \frac{\pi}{4}$. Find the value of $|\cot B - \cot C|$.

2. In a $\Delta ABC, a = \sqrt{3}, b = 3$ and $\angle C = \frac{\pi}{3}$. Let internal angle bisector of angle C intersects side AB at D and altitude from B meets the angle bisector CD at E. If O_1 and O_2 are incentres of ΔBEC and ΔBED. Find the distance between the vertex B and orthocentre of $\Delta O_1 EO_2$.

3. In a ΔABC ; inscribed circle with centre I touches sides AB, AC and BC at D, E, F respectively. Let area of quadrilateral $ADIE$ is 5 units and area of quadrilteral $BFID$ is 10 units. Find the value of $\dfrac{\cos\left(\frac{C}{2}\right)}{\sin\left(\frac{A-B}{2}\right)}$.

4. If Δ be area of incircle of a triangle ABC and $\Delta_1, \Delta_2, \Delta_3$ be the area of excircles then find the least value of $\dfrac{\Delta_1\Delta_2\Delta_3}{729\,\Delta^3}$.

5. In ΔABC, $b = c$, $\angle A = 106°, M$ is an interior point such that $\angle MBA = 7°$, $\angle MAB = 23°$ and $\angle MCA = \theta°$, then $\frac{\theta}{2}$ is equal to

(where notations have their usual meaning)

6. In an acute angled triangle ABC, $\angle A = 20°$, let DEF be the feet of altitudes through A, B, C respectively and H is the orthocentre of ΔABC. Find $\dfrac{AH}{AD} + \dfrac{BH}{BE} + \dfrac{CH}{CF}$.

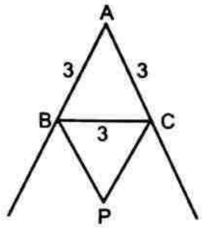
7. Let ΔABC be inscribed in a circle having radius unity. The three internal bisectors of the angles A, B and C are extended to intersect the circumcircle of ΔABC at A_1, B_1 and C_1 respectively. Then $\dfrac{AA_1 \cos\frac{A}{2} + BB_1 \cos\frac{B}{2} + CC_1 \cos\frac{C}{2}}{\sin A + \sin B + \sin C} =$

8. If the quadratic equation $ax^2 + bx + c = 0$ has equal roots where a, b, c denotes the lengths of the sides opposite to vertex A, B and C of the ΔABC respectively. Find the number of integers in the range of $\dfrac{\sin A}{\sin C} + \dfrac{\sin C}{\sin A}$.

9. If in the triangle ABC, $\tan\frac{A}{2}, \tan\frac{B}{2}$ and $\tan\frac{C}{2}$ are in harmonic progression then the least value of $\cot^2\frac{B}{2}$ is equal to :

10. In ΔABC, if circumradius 'R' and inradius 'r' are connected by relation $R^2 - 4Rr + 8r^2 - 12r + 9 = 0$, then the greatest integer which is less than the semiperimeter of ΔABC is :

11. Sides AB and AC in an equilateral triangle ABC with side length 3 is extended to form two rays from point A as shown in the figure. Point P is chosen outside the triangle ABC and between the two rays such that $\angle ABP + \angle BCP = 180°$. If the maximum length of CP is M, then $M^2/2$ is equal to :



12. Let a, b, c be sides of a triangle ABC and Δ denotes its area.
If $a = 2$; $\Delta = \sqrt{3}$ and $a\cos C + \sqrt{3}\, a\sin C - b - c = 0$; then find the value of $(b + c)$.
(symbols used have usual meaning in ΔABC).

13. If circumradius of ΔABC is 3 units and its area is 6 units and ΔDEF is formed by joining foot of perpendiculars drawn from A, B, C on sides BC, CA, AB respectively. Find the perimeter of ΔDEF.

Answers

1.	2	**2.**	1	**3.**	3	**4.**	1	**5.**	7	**6.**	2	**7.**	2
8.	3	**9.**	3	**10.**	7	**11.**	6	**12.**	4	**13.**	4		

Chapter 25 – Inverse Trigonometric Functions

25 INVERSE TRIGONOMETRIC FUNTIONS

Exercise-1 : Single Choice Problems

1. If $\sin^{-1} x \in \left(0, \frac{\pi}{2}\right)$, then the value of $\tan\left(\frac{\cos^{-1}(\sin(\cos^{-1} x)) + \sin^{-1}(\cos(\sin^{-1} x))}{2}\right)$ is :

(a) 1 (b) 2 (c) 3 (d) 4

2. The solution set of $(\cot^{-1} x)(\tan^{-1} x) + \left(2 - \frac{\pi}{2}\right)\cot^{-1} x - 3\tan^{-1} x - 3\left(2 - \frac{\pi}{2}\right) > 0$, is :

(a) $x \in (\tan 2, \tan 3)$ (b) $x \in (\cot 3, \cot 2)$

(c) $x \in (-\infty, \tan 2) \cup (\tan 3, \infty)$ (d) $x \in (-\infty, \cot 3) \cup (\cot 2, \infty)$

3. The value of $\sec^2(\tan^{-1} 2) + \text{cosec}^2(\cot^{-1} 3)$ is :

(a) 14 (b) 15 (c) 16 (d) 17

4. Sum the series :

$\tan^{-1}\left(\frac{4}{1+3\cdot 4}\right) + \tan^{-1}\left(\frac{6}{1+8\cdot 9}\right) + \tan^{-1}\left(\frac{8}{1+15\cdot 16}\right) + \ldots\ldots \infty$ is :

(a) $\cot^{-1}(2)$ (b) $\tan^{-1}(2)$ (c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

5. $\cot^{-1}(\sqrt{\cos\alpha}) - \tan^{-1}(\sqrt{\cos\alpha}) = x$, then $\sin x =$

(a) $\tan^2\left(\frac{\alpha}{2}\right)$ (b) $\cot^2\left(\frac{\alpha}{2}\right)$ (c) $\tan\alpha$ (d) $\cot\left(\frac{\alpha}{2}\right)$

6. The sum of the infinite series $\cot^{-1}\left(\frac{7}{4}\right) + \cot^{-1}\left(\frac{19}{4}\right) + \cot^{-1}\left(\frac{39}{4}\right) + \cot^{-1}\left(\frac{67}{4}\right) + \ldots\ldots \infty$ is :

(a) $\frac{\pi}{4} - \cot^{-1}(3)$ (b) $\frac{\pi}{4} - \tan^{-1}(3)$ (c) $\frac{\pi}{4} + \cot^{-1}(3)$ (d) $\frac{\pi}{4} + \tan^{-1}(3)$

7. The number of solutions of equation $\cos^{-1}(1-x) + m\cos^{-1} x = \frac{n\pi}{2}$ is : (where $m > 0; n \leq 0$)

(a) 0 (b) 1 (c) 2 (d) none of these

8. Number of solution(s) of the equation $2\tan^{-1}(2x-1) = \cos^{-1}(x)$ is :

(a) 1 (b) 2 (c) 3 (d) infinitely many

9. $\sin^{-1}\left(\frac{x^2}{4} + \frac{y^2}{9}\right) + \cos^{-1}\left(\frac{x}{2\sqrt{2}} + \frac{y}{3\sqrt{2}} - 2\right)$ equals to :

(a) $\frac{\pi}{2}$ (b) π (c) $\frac{\pi}{\sqrt{2}}$ (d) $\frac{3\pi}{2}$

10. The complete solution set of the inequality $(\cos^{-1} x)^2 - (\sin^{-1} x)^2 > 0$ is :

(a) $\left[0, \frac{1}{\sqrt{2}}\right)$ (b) $\left[-1, \frac{1}{\sqrt{2}}\right)$ (c) $(-1, 1)$ (d) $\left[-1, \frac{1}{2}\right)$

11. Let α, β are the roots of the equation $x^2 + 7x + k(k-3) = 0$, where $k \in (0, 3)$ and k is a constant. Then the value of $\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\frac{1}{\alpha} + \tan^{-1}\frac{1}{\beta}$ is :

(a) π (b) $\frac{\pi}{2}$ (c) 0 (d) $-\frac{\pi}{2}$

12. Let $f(x) = a + 2b\cos^{-1} x$, $b > 0$. If domain and range of $f(x)$ are the same set, then $(b-a)$ is equal to :

(a) $1 - \frac{1}{\pi}$ (b) $\frac{2}{\pi}$

(c) $\frac{2}{\pi} + 1$ (d) $1 + \frac{1}{\pi}$

13. If $(\tan^{-1} x)^2 + (\cot^{-1} x)^2 = \frac{5\pi^2}{8}$, then x equals to :

(a) -1 (b) 1 (c) 0 (d) $\sqrt{3}$

14. The total number of ordered pairs (x, y) satisfying $|y| = \cos x$ and $y = \sin^{-1}(\sin x)$, where $x \in [-2\pi, 3\pi]$ is equal to :

(a) 2 (b) 4 (c) 5 (d) 6

15. If $[\sin^{-1}(\cos^{-1}(\sin^{-1}(\tan^{-1} x)))] = 1$, where $[\cdot]$ denotes greatest integer function, then complete set of values of x is :

(a) $[\tan(\sin(\cos 1)), \tan(\cos(\sin 1))]$ (b) $[\tan(\sin(\cos 1)), \tan(\sin(\cos(\sin 1)))]$

(c) $[\tan(\cos(\sin 1)), \tan(\sin(\cos(\sin 1)))]$ (d) $[\tan(\sin(\cos 1)), 1]$

16. The number of ordered pair(s) (x, y) of real numbers satisfying the equation $1 + x^2 + 2x\sin(\cos^{-1} y) = 0$, is :

(a) 0 (b) 1 (c) 2 (d) 3

17. The value of $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$ is :

(a) $\frac{\pi}{2}$ (b) π (c) $\frac{3\pi}{4}$ (d) $\frac{5\pi}{8}$

18. The complete set of values of x for which $2\tan^{-1}x + \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is independent of x is :

(a) $(-\infty, 0]$ (b) $[0, \infty)$ (c) $(-\infty, -1]$ (d) $[1, \infty)$

19. The number of ordered pair(s) (x, y) which satisfy $y = \tan^{-1}\tan x$ and $16(x^2+y^2) - 48\pi x + 16\pi y + 31\pi^2 = 0$, is :

(a) 0 (b) 1 (c) 2 (d) 3

20. Domain (D) and range (R) of $f(x) = \sin^{-1}(\cos^{-1}[x])$ where [ ] denotes the greatest integer function is

(a) $D \equiv [1, 2), R \equiv \{0\}$ (b) $D \equiv [0, 1), R \equiv \{-1, 0, 1\}$

(c) $D \equiv [-1, 1), R \equiv \left\{0, \frac{\pi}{2}, \pi\right\}$ (d) $D \equiv [-1, 1), R \equiv \left\{-\frac{\pi}{2}, 0, \frac{\pi}{2}\right\}$

21. If $2\sin^{-1}x + \{\cos^{-1}x\} > \frac{\pi}{2} + \{\sin^{-1}x\}$, then $x \in$: (where $\{\cdot\}$ denotes fractional part function)

(a) $(\cos 1, 1]$ (b) $[\sin 1, 1]$ (c) $(\sin 1, 1]$ (d) None of these

22. Let $f(x) = x^{11} + x^9 - x^7 + x^3 + 1$ and $f(\sin^{-1}(\sin 8)) = \alpha$, ($\alpha$ is constant). If $f(\tan^{-1}(\tan 8)) = \lambda - \alpha$, then the value of λ is :

(a) 2 (b) 3 (c) 4 (d) 1

23. The number of real values of x satisfying the equation $3\sin^{-1}x + \pi x - \pi = 0$ is/are :

(a) 0 (b) 1 (c) 2 (d) 3

24. Range of $f(x) = \sin^{-1}x + x^2 + 4x + 1$ is :

(a) $\left[-\frac{\pi}{2} - 2, \frac{\pi}{2} + 6\right]$ (b) $\left[0, \frac{\pi}{2} + 6\right]$ (c) $\left[-\frac{\pi}{2} - 2, \infty\right)$ (d) $[-3, \infty)$

25. The solution set of the inequality $(\text{cosec}^{-1}x)^2 - 2\text{cosec}^{-1}x \ge \frac{\pi}{6}(\text{cosec}^{-1}x - 2)$ is $(-\infty, a] \cup [b, \infty)$, then $(a+b)$ equals :

(a) 0 (b) 1 (c) 2 (d) -3

26. Number of solution of the equation $2\sin^{-1}(x+2) = \cos^{-1}(x+3)$ is :

(a) 0 (b) 1 (c) 2 (d) None of these

27. $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + \ldots\ldots \infty =$

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

28. If $\tan^{-1}\frac{1}{4} + \tan^{-1}\frac{2}{9} = \frac{1}{2}\cos^{-1}x$ then x is equal to :

(a) $\frac{1}{2}$ (b) $\frac{2}{5}$ (c) $\frac{3}{5}$ (d) none of these

29. The set of value of x, satisfying the equation $\tan^2(\sin^{-1} x) > 1$ is :

(a) $(-1, 1)$
(b) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
(c) $[-1, 1]-\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
(d) $(-1, 1)-\left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

30. The sum of the series $\cot^{-1}\left(\frac{9}{2}\right)+\cot^{-1}\left(\frac{33}{4}\right)+\cot^{-1}\left(\frac{129}{8}\right)+\ldots\ldots \infty$ is equal to :

(a) $\cot^{-1}(2)$ (b) $\cot^{-1}(3)$ (c) $\cot^{-1}(-1)$ (d) $\cot^{-1}(1)$

31. If $\int \frac{\ln(\cot x)}{\sin x \cos x} dx = -\frac{1}{k}\ln^2(\cot x)+C$

(where C is a constant); then the value of k is :

(a) 1 (b) 2 (c) 3 (d) $\frac{1}{2}$

32. The number of solutions of $\sin^{-1} x+\sin^{-1}(1+x)=\cos^{-1} x$ is/are :

(a) 0 (b) 1 (c) 2 (d) infinite

33. The value of x satisfying the equation

$$(\sin^{-1} x)^3-(\cos^{-1} x)^3+(\sin^{-1} x)(\sin^{-1} x-\cos^{-1} x)=\frac{\pi^3}{16} \text{ is :}$$

(a) $\cos\frac{\pi}{5}$ (b) $\cos\frac{\pi}{4}$ (c) $\cos\frac{\pi}{8}$ (d) $\cos\frac{\pi}{12}$

34. The complete solution set of the equation

$$\sin^{-1}\sqrt{\frac{1+x}{2}}-\sqrt{2-x}=\cot^{-1}(\tan\sqrt{2-x})-\sin^{-1}\sqrt{\frac{1-x}{2}} \text{ is :}$$

(a) $\left[2-\frac{\pi^2}{4}, 1\right]$ (b) $\left[1-\frac{\pi^2}{4}, 1\right]$ (c) $\left[2-\frac{\pi^2}{4}, 0\right]$ (d) $[-1, 1]$

35. Let $f(x)=\tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$ then which of the following is correct :

(a) $f(x)$ has only one integer in its range
(b) Range of $f(x)$ is $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)-\{0\}$
(c) Range of $f(x)$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)-\{0\}$
(d) Range of $f(x)$ is $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]-\{0\}$

36. If $\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{2}{9}=\frac{1}{2}\cos^{-1} x$ then x is equal to :

(a) $\frac{1}{2}$ (b) $\frac{2}{5}$ (c) $\frac{3}{5}$ (d) None of these

37. The set of values of x, satisfying the equation $\tan^2(\sin^{-1} x) > 1$ is :

(a) $(-1, 1)$
(b) $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
(c) $[-1, 1] - \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
(d) $(-1, 1) - \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$

38. The sum of the series $\cot^{-1}\left(\frac{9}{2}\right) + \cot^{-1}\left(\frac{33}{4}\right) + \cot^{-1}\left(\frac{129}{8}\right) + \ldots\ldots \infty$ is equal to

(a) $\cot^{-1}(2)$
(b) $\cot^{-1}(3)$
(c) $\cot^{-1}(-1)$
(d) $\cot^{-1}(1)$

39. The number of real values of x satisfying $\tan^{-1}\left(\frac{x}{1-x^2}\right) + \tan^{-1}\left(\frac{1}{x^3}\right) = \frac{3\pi}{4}$ is :

(a) 0
(b) 1
(c) 2
(d) infinitely many

40. Number of integral values of λ such that the equation $\cos^{-1} x + \cot^{-1} x = \lambda$ possesses solution is:

(a) 2
(b) 8
(c) 5
(d) 10

41. If the equation $x^3 + bx^2 + cx + 1 = 0\,(b < c)$ has only one real root α.
Then the value of $2\tan^{-1}(\operatorname{cosec}\alpha) + \tan^{-1}(2\sin\alpha\sec^2\alpha)$ is :

(a) $-\frac{\pi}{2}$
(b) $-\pi$
(c) $\frac{\pi}{2}$
(d) π

42. Range of the function $f(x) = \cot^{-1}\{-x\} + \sin^{-1}\{x\} + \cos^{-1}\{x\}$, where $\{\cdot\}$ denotes fractional part function

(a) $\left(\frac{3\pi}{4}, \pi\right)$
(b) $\left[\frac{3\pi}{4}, \pi\right)$
(c) $\left[\frac{3\pi}{4}, \pi\right]$
(d) $\left(\frac{3\pi}{4}, \pi\right]$

43. If $3 \le a < 4$ then the value of $\sin^{-1}(\sin[a]) + \tan^{-1}(\tan[a]) + \sec^{-1}(\sec[a])$, where $[x]$ denotes greatest integer function less than or equal to x, is equal to :

(a) 3
(b) $2\pi - 9$
(c) $2\pi - 3$
(d) $9 - 2\pi$

44. The number of real solutions of $y + y^2 = \sin x$ and $y + y^3 = \cos^{-1}\cos x$ is/are

(a) 0
(b) 1
(c) 3
(d) Infinite

45. Range of $f(x) = \sin^{-1}[x-1] + 2\cos^{-1}[x-2]$ ($[\cdot]$ denotes greatest integer function)

(a) $\left\{-\frac{\pi}{2}, 0\right\}$
(b) $\left\{\frac{\pi}{2}, 2\pi\right\}$
(c) $\left\{\frac{\pi}{4}, \frac{\pi}{2}\right\}$
(d) $\left\{\frac{3\pi}{2}, 2\pi\right\}$

Answers

1.	(a)	**2.**	(b)	**3.**	(b)	**4.**	(a)	**5.**	(a)	**6.**	(c)	**7.**	(a)	**8.**	(a)	**9.**	(d)	**10.**	(b)
11.	(c)	**12.**	(d)	**13.**	(a)	**14.**	(c)	**15.**	(b)	**16.**	(b)	**17.**	(b)	**18.**	(a)	**19.**	(d)	**20.**	(a)
21.	(b)	**22.**	(a)	**23.**	(b)	**24.**	(a)	**25.**	(b)	**26.**	(b)	**27.**	(a)	**28.**	(c)	**29.**	(d)	**30.**	(a)
31.	(b)	**32.**	(b)	**33.**	(c)	**34.**	(a)	**35.**	(b)	**36.**	(c)	**37.**	(d)	**38.**	(a)	**39.**	(a)	**40.**	(c)
41.	(b)	**42.**	(d)	**43.**	(a)	**44.**	(d)	**45.**	(d)										

Exercise-2 : One or More than One Answer is/are Correct

1. $f(x) = \sin^{-1}(\sin x), g(x) = \cos^{-1}(\cos x)$, then :

(a) $f(x) = g(x)$ if $x \in \left(0, \frac{\pi}{4}\right)$
(b) $f(x) < g(x)$ if $x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$
(c) $f(x) < g(x)$ if $\left(\pi, \frac{5\pi}{4}\right)$
(d) $f(x) > g(x)$ if $x \in \left(\pi, \frac{5\pi}{4}\right)$

2. The solution(s) of the equation $\cos^{-1} x = \tan^{-1} x$ satisfy

(a) $x^2 = \frac{\sqrt{5}-1}{2}$
(b) $x^2 = \frac{\sqrt{5}+1}{2}$
(c) $\sin(\cos^{-1} x) = \frac{\sqrt{5}-1}{2}$
(d) $\tan(\cos^{-1} x) = \frac{\sqrt{5}-1}{2}$

3. If the numerical value of $\tan\left(\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(\frac{2}{3}\right)\right)$ is $\left(\frac{a}{b}\right)$, where a, b are two positive integers and their H.C.F is 1

(a) $a + b = 23$
(b) $a - b = 11$
(c) $3b = a + 1$
(d) $2a = 3b$

4. A solution of the equation $\cot^{-1} 2 = \cot^{-1} x + \cot^{-1}(10 - x)$ where $1 < x < 9$ is :

(a) 7
(b) 3
(c) 2
(d) 5

5. Consider the equation $\sin^{-1}\left(x^2 - 6x + \frac{17}{2}\right) + \cos^{-1} k = \frac{\pi}{2}$, then :

(a) the largest value of k for which equation has 2 distinct solution is 1
(b) the equation must have real root if $k \in \left(-\frac{1}{2}, 1\right)$
(c) the equation must have real root if $k \in \left(-1, \frac{1}{2}\right)$
(d) the equation has unique solution if $k = -\frac{1}{2}$

6. The value of x satisfying the equation

$$(\sin^{-1} x)^3 - (\cos^{-1} x)^3 + (\sin^{-1} x)(\cos^{-1} x)(\sin^{-1} x - \cos^{-1} x) = \frac{\pi^3}{16}$$

can not be equal to :

(a) $\cos\frac{\pi}{5}$
(b) $\cos\frac{\pi}{4}$
(c) $\cos\frac{\pi}{8}$
(d) $\cos\frac{\pi}{12}$

Answers

1.	(a, b, c)	2.	(a, c)	3.	(a, b, c)	4.	(a, b)	5.	(a, b, d)	6.	(a, b, d)

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 2

Let $\cos^{-1}(4x^3 - 3x) = a + b\cos^{-1} x$

1. If $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$, then $\sin^{-1}\left(\sin \frac{a}{b}\right)$ is :

(a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$ (c) $-\frac{\pi}{6}$ (d) $\frac{\pi}{6}$

2. If $x \in \left(\frac{1}{2}, 1\right]$, then $\lim\limits_{y \to a} b\cos y$ is :

(a) $-\frac{1}{3}$ (b) -3 (c) $\frac{1}{3}$ (d) 3

Answers

1.	(a)	**2.**	(d)																

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	$\sin^{-1}\frac{4}{5}+2\tan^{-1}\frac{1}{3}=$	(P)	$\frac{\pi}{6}$
(B)	$\sin^{-1}\frac{12}{13}+\cos^{-1}\frac{4}{5}+\tan^{-1}\frac{63}{16}=$	(Q)	$\frac{\pi}{2}$
(C)	If $A=\tan^{-1}\frac{x\sqrt{3}}{2\lambda-x}$, $B=\tan^{-1}\left(\frac{2x-\lambda}{\lambda\sqrt{3}}\right)$ then $A-B$ can be equal to	(R)	$\frac{\pi}{4}$
(D)	$\tan^{-1}\frac{1}{7}+2\tan^{-1}\frac{1}{3}=$	(S)	π
		(T)	$\frac{\pi}{3}$

2.

	Column-I		Column-II
(P)	If $f(x)=\sin^{-1}x$ and $\lim\limits_{x\to\frac{1}{2}^+} f(3x-4x^3)=l-3\left(\lim\limits_{x\to\frac{1}{2}^+} f(x)\right)$ then $[l]=$ ([·] denotes greatest integer function)	(P)	3
(Q)	If $x>1$, then the value of $\sin\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2}-\tan^{-1}x\right)$ is	(Q)	-1
(R)	Number of values of x satisfying $\sin^{-1}x-\cos^{-1}x=\sin^{-1}(3x-2)$	(R)	2
(S)	The value of $\sin\left(\tan^{-1}3+\tan^{-1}\frac{1}{3}\right)$	(S)	1

3.

	Column-I		Column-II
(A)	If the first term of an arithmetic progression is 1, its second term is n, and the sum of the first n terms is $33n$	(P)	3
(B)	If the equation $\cos^{-1} x + \cot^{-1} x = k$ possess solution, then the largest integral value of k is	(Q)	4
(C)	The number of solution of equation $\cos\theta = \lvert 1 + \sin\theta \rvert$ in interval $[0, 3\pi]$, is	(R)	5
(D)	If the quadratic equation $x^2 - x - a = 0$ has integral roots where $a \in N$ and $4 \le a \le 40$, then the number of possible values of a is	(S)	9

4.

	Column-I		Column-II
(A)	The value of $\tan^{-1}([\pi]) + \tan^{-1}([-\pi] + 1) =$ ([·] denotes greatest integer function)	(P)	2
(B)	The number of solutions of the equation $\tan x + \sec x = 2\cos x$ in the interval $[0, 2\pi]$ is	(Q)	3
(C)	The number of roots of the equation $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is	(R)	0
(D)	The number of solutions of the equation $x^3 + x^2 + 4x + 2\sin x = 0$ in the interval $[0, 2\pi]$ is	(S)	1

Answers

1. A → Q ; B → S ; C → P ; D → R

2. A → P ; B → Q ; C → R ; D → S

3. A → S ; B → R ; C → P ; D → Q

4. A → R ; B → P ; C → Q ; D → S

Exercise-5 : Subjective Type Problems

1. The complete set of values of x satisfying the inequality $\sin^{-1}(\sin 5) > x^2 - 4x$ is $(2-\sqrt{\lambda-2\pi}, 2+\sqrt{\lambda-2\pi})$, then $\lambda =$
2. In a ΔABC ; if $(II_1)^2 + (I_2I_3)^2 = \lambda R^2$, where I denotes incentre ; I_1, I_2 and I_3 denote centres of the circles escribed to the sides BC, CA and AB respectively and R be the radius of the circum circle of ΔABC. Find λ.
3. If $2\tan^{-1}\dfrac{1}{5} - \sin^{-1}\dfrac{3}{5} = -\cos^{-1}\dfrac{63}{\lambda}$, then $\lambda =$
4. If $2\tan^{-1}\dfrac{1}{5} - \sin^{-1}\dfrac{3}{5} = -\cos^{-1}\dfrac{9\lambda}{65}$, then $\lambda =$
5. If $\sum_{n=0}^{\infty} 2\cot^{-1}\left(\dfrac{n^2+n+4}{2}\right) = k\pi$, then find the value of k.
6. Find number of solutions of the equation $\sin^{-1}(|\log_6^2(\cos x) - 1|) + \cos^{-1}(|3\log_6^2(\cos x) - 7|) = \dfrac{\pi}{2}$, if $x \in [0, 4\pi]$.

Answers

1.	9	**2.**	16	**3.**	65	**4.**	7	**5.**	1	**6.**	4

□□□

Chapter 26 – Vector & 3Dimensional Geometry

Vector & 3Dimensional Geometry

26 VECTOR & 3DIMENSIONAL GEOMETRY

Exercise-1 : Single Choice Problems

1. The minimum value of $x^2 + y^2 + z^2$ if $ax + by + cz = p$, is :

(a) $\left(\dfrac{p}{a+b+c}\right)^2$ (b) $\dfrac{p^2}{a^2+b^2+c^2}$ (c) $\dfrac{a^2+b^2+c^2}{p^2}$ (d) 0

2. If the angle between the vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is $\dfrac{\pi}{3}$ and the area of the triangle with adjacent sides equal to $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is 3, then $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}}$ is equal to :

(a) $\sqrt{3}$ (b) $2\sqrt{3}$ (c) $4\sqrt{3}$ (d) $\dfrac{\sqrt{3}}{2}$

3. A straight line cuts the sides AB, AC and AD of a parallelogram $ABCD$ at points B_1, C_1 and D_1 respectively. If $\overrightarrow{\mathbf{AB_1}} = \lambda_1 \overrightarrow{\mathbf{AB}}, \overrightarrow{\mathbf{AD_1}} = \lambda_2 \overrightarrow{\mathbf{AD}}$ and $\overrightarrow{\mathbf{AC_1}} = \dfrac{\lambda_3}{2} \overrightarrow{\mathbf{AC}}$, where λ_1, λ_2 and λ_3 are positive real numbers, then :

(a) λ_1, λ_3 and λ_2 are in AP
(b) λ_1, λ_3 and λ_2 are in GP
(c) λ_1, λ_3 and λ_2 are in HP
(d) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

4. Let $\vec{\mathbf{a}} = 2\hat{\mathbf{i}} + \hat{\mathbf{j}} - 2\hat{\mathbf{k}}$ and $\vec{\mathbf{b}} = \hat{\mathbf{i}} + \hat{\mathbf{j}}$. If $\vec{\mathbf{c}}$ is a vector such that $\vec{\mathbf{a}} \cdot \vec{\mathbf{c}} = |\vec{\mathbf{c}}|, |\vec{\mathbf{c}} - \vec{\mathbf{a}}| = 2\sqrt{2}$ and the angle between $\vec{\mathbf{a}} \times \vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ is 30° then $\left|(\vec{\mathbf{a}} \times \vec{\mathbf{b}}) \times \vec{\mathbf{c}}\right|$ is equal to :

(a) $\dfrac{2}{3}$ (b) $\dfrac{3}{2}$ (c) 2 (d) 3

5. If acute angle between the line $\vec{\mathbf{r}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \lambda(4\hat{\mathbf{i}} - 3\hat{\mathbf{k}})$ and xy-plane is θ_1 and acute angle between the planes $x + 2y = 0$ and $2x + y = 0$ is θ_2, then $(\cos^2\theta_1 + \sin^2\theta_2)$ equals to :

(a) 1 (b) $\dfrac{1}{4}$ (c) $\dfrac{2}{3}$ (d) $\dfrac{3}{4}$

6. If a, b, c, x, y, z are real and $a^2 + b^2 + c^2 = 25$, $x^2 + y^2 + z^2 = 36$ and $ax + by + cz = 30$, then $\dfrac{a+b+c}{x+y+z}$ is equal to :

(a) 1 (b) $\dfrac{6}{5}$ (c) $\dfrac{5}{6}$ (d) $\dfrac{3}{4}$

7. If $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are non-zero, non-collinear vectors such that $|\vec{\mathbf{a}}| = 2$, $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = 1$ and angle between $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is $\frac{\pi}{3}$. If $\vec{\mathbf{r}}$ is any vector such that $\vec{\mathbf{r}} \cdot \vec{\mathbf{a}} = 2$, $\vec{\mathbf{r}} \cdot \vec{\mathbf{b}} = 8$, $(\vec{\mathbf{r}} + 2\vec{\mathbf{a}} - 10\vec{\mathbf{b}}) \cdot (\vec{\mathbf{a}} \times \vec{\mathbf{b}}) = 4\sqrt{3}$ and satisfy to $\vec{\mathbf{r}} + 2\vec{\mathbf{a}} - 10\vec{\mathbf{b}} = \lambda(\vec{\mathbf{a}} \times \vec{\mathbf{b}})$, then λ is equal to :

(a) $\dfrac{1}{2}$ (b) 2 (c) $\dfrac{1}{4}$ (d) None of these

8. Let $\vec{\mathbf{a}} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$; $\vec{\mathbf{b}} = 2(\hat{\mathbf{i}} + \hat{\mathbf{k}})$ and $\vec{\mathbf{c}} = 4\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}$. Sum of the values of '$\alpha$' for which the equation $x\,\vec{\mathbf{a}} + y\,\vec{\mathbf{b}} + z\,\vec{\mathbf{c}} = \alpha(x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}})$ has non-trivial solution is :

(a) -1 (b) 4 (c) 7 (d) 8

9. If $\vec{\mathbf{a}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\vec{\mathbf{b}} = \hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}$, $\vec{\mathbf{c}} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$, then the value of $\begin{vmatrix} \vec{\mathbf{a}} \cdot \vec{\mathbf{a}} & \vec{\mathbf{a}} \cdot \vec{\mathbf{b}} & \vec{\mathbf{a}} \cdot \vec{\mathbf{c}} \\ \vec{\mathbf{b}} \cdot \vec{\mathbf{a}} & \vec{\mathbf{b}} \cdot \vec{\mathbf{b}} & \vec{\mathbf{b}} \cdot \vec{\mathbf{c}} \\ \vec{\mathbf{c}} \cdot \vec{\mathbf{a}} & \vec{\mathbf{c}} \cdot \vec{\mathbf{b}} & \vec{\mathbf{c}} \cdot \vec{\mathbf{c}} \end{vmatrix}$ is equal to :

(a) 2 (b) 4 (c) 16 (d) 64

10. If $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are two vectors such that $|\vec{\mathbf{a}}| = 1$, $|\vec{\mathbf{b}}| = 4$, $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} = 2$. If $\vec{\mathbf{c}} = (2\vec{\mathbf{a}} \times \vec{\mathbf{b}}) - 3\vec{\mathbf{b}}$, then angle between $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ is :

(a) $\dfrac{\pi}{6}$ (b) $\dfrac{\pi}{3}$ (c) $\dfrac{2\pi}{3}$ (d) $\dfrac{5\pi}{6}$

11. If $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ are unit vectors, then the value of $|\vec{\mathbf{a}} - 2\vec{\mathbf{b}}|^2 + |\vec{\mathbf{b}} - 2\vec{\mathbf{c}}|^2 + |\vec{\mathbf{c}} - 2\vec{\mathbf{a}}|^2$ does not exceed to:

(a) 9 (b) 12 (c) 18 (d) 21

12. The adjacent side vectors $\overrightarrow{\mathbf{OA}}$ and $\overrightarrow{\mathbf{OB}}$ of a rectangle $OACB$ are $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ respectively, where O is the origin. If $16|\vec{\mathbf{a}} \times \vec{\mathbf{b}}| = 3(|\vec{\mathbf{a}}| + |\vec{\mathbf{b}}|)^2$ and θ be the acute angle between the diagonals OC and AB then the value of $\tan(\theta/2)$ is :

(a) $\dfrac{1}{\sqrt{2}}$ (b) $\dfrac{1}{2}$ (c) $\dfrac{1}{\sqrt{3}}$ (d) $\dfrac{1}{3}$

13. The vector $\overrightarrow{\mathbf{AB}} = 3\hat{\mathbf{i}} + 4\hat{\mathbf{k}}$ and $\overrightarrow{\mathbf{AC}} = 5\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 4\hat{\mathbf{k}}$ are the sides of a triangle ABC. The length of the median through A is :

(a) $\sqrt{288}$ (b) $\sqrt{72}$ (c) $\sqrt{33}$ (d) $\sqrt{18}$

14. If $\vec{a} = 2\hat{i} + \lambda\hat{j} + 3\hat{k}$; $\vec{b} = 3\hat{i} + 3\hat{j} + 5\hat{k}$; $\vec{c} = \lambda\hat{i} + 2\hat{j} + 2\hat{k}$ are linearly dependent vectors, then the number of possible values of λ is :

(a) 0 (b) 1 (c) 2 (d) More than 2

15. The scalar triple product $[\vec{a}+\vec{b}-\vec{c}\ \ \vec{b}+\vec{c}-\vec{a}\ \ \vec{c}+\vec{a}-\vec{b}]$ is equal to :

(a) 0 (b) $[\vec{a}\,\vec{b}\,\vec{c}]$ (c) $2[\vec{a}\,\vec{b}\,\vec{c}]$ (d) $4[\vec{a}\,\vec{b}\,\vec{c}]$

16. If $\hat{a}$ and $\hat{b}$ are unit vectors then the vector defined as $\vec{V} = (\hat{a} \times \hat{b}) \times (\hat{a} + \hat{b})$ is collinear to the vector :

(a) $\hat{a} + \hat{b}$ (b) $\hat{b} - \hat{a}$ (c) $2\hat{a} - \hat{b}$ (d) $\hat{a} + 2\hat{b}$

17. The sine of angle formed by the lateral face *ADC* and plane of the base *ABC* of the tetrahedron *ABCD*, where $A \equiv (3, -2, 1)$; $B \equiv (3, 1, 5)$; $C \equiv (4, 0, 3)$ and $D \equiv (1, 0, 0)$, is :

(a) $\dfrac{2}{\sqrt{29}}$ (b) $\dfrac{5}{\sqrt{29}}$ (c) $\dfrac{3\sqrt{3}}{\sqrt{29}}$ (d) $\dfrac{-2}{\sqrt{29}}$

18. Let $\vec{a}_r = x_r\hat{i} + y_r\hat{j} + z_r\hat{k}$, $r = 1, 2, 3$ be three mutually perpendicular unit vectors, then the value of $\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix}$ is equal to :

(a) 0 (b) ± 1 (c) ± 2 (d) ± 4

19. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors and $\vec{r}$ be any arbitrary vector, then the expression $(\vec{a}\times\vec{b})\times(\vec{r}\times\vec{c}) + (\vec{b}\times\vec{c})\times(\vec{r}\times\vec{a}) + (\vec{c}\times\vec{a})\times(\vec{r}\times\vec{b})$ is always equal to :

(a) $[\vec{a}\,\vec{b}\,\vec{c}]\,\vec{r}$ (b) $2[\vec{a}\,\vec{b}\,\vec{c}]\,\vec{r}$ (c) $4[\vec{a}\,\vec{b}\,\vec{c}]\,\vec{r}$ (d) $\vec{0}$

20. *E* and *F* are the interior points on the sides *BC* and *CD* of a parallelogram *ABCD*. Let $\overrightarrow{BE} = 4\,\overrightarrow{EC}$ and $\overrightarrow{CF} = 4\,\overrightarrow{FD}$. If the line *EF* meets the diagonal *AC* in *G*, then $\overrightarrow{AG} = \lambda\,\overrightarrow{AC}$, where λ is equal to :

(a) $\dfrac{1}{3}$ (b) $\dfrac{21}{25}$ (c) $\dfrac{7}{13}$ (d) $\dfrac{21}{5}$

21. If $\hat{a}, \hat{b}$ are unit vectors and $\vec{c}$ is such that $\vec{c} = \vec{a} \times \vec{c} + \vec{b}$, then the maximum value of $[\vec{a}\,\vec{b}\,\vec{c}]$ is :

(a) 1 (b) $\dfrac{1}{2}$ (c) 2 (d) $\dfrac{3}{2}$

22. Consider matrices $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & -1 \\ 2 & 2 & 3 \end{bmatrix}$; $C = \begin{bmatrix} 14 \\ 12 \\ 2 \end{bmatrix}$; $D = \begin{bmatrix} 13 \\ 11 \\ 14 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that solutions of equation $AX = C$ and $BX = D$ represents two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ respectively in three dimensional space. If $P'Q'$ is the reflection of the line PQ in the plane $\prod : x + y + z = 9$, then the point which does not lie on $P'Q'$ is :

(a) (3, 4, 2) (b) (5, 3, 4) (c) (7, 2, 3) (d) (1, 5, 6)

23. The value of α for which point $M(\alpha\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$, lies in the plane containing three points $A(\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$, $B(2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and $C(3\hat{\mathbf{i}} - \hat{\mathbf{k}})$ is :

(a) 1 (b) 2 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

24. Q is the image of point $P(1, -2, 3)$ with respect to the plane $x - y + z = 7$. The distance of Q from the origin is :

(a) $\sqrt{\frac{70}{3}}$ (b) $\frac{1}{2}\sqrt{\frac{70}{3}}$ (c) $\sqrt{\frac{35}{3}}$ (d) $\sqrt{\frac{15}{2}}$

25. $\hat{\mathbf{a}}, \hat{\mathbf{b}}$ and $\hat{\mathbf{a}} - \hat{\mathbf{b}}$ are unit vectors. The volume of the parallelopiped, formed with $\hat{\mathbf{a}}, \hat{\mathbf{b}}$ and $\hat{\mathbf{a}} \times \hat{\mathbf{b}}$ as coterminous edges is :

(a) 1 (b) $\frac{1}{4}$ (c) $\frac{2}{3}$ (d) $\frac{3}{4}$

26. A line passing through $P(3, 7, 1)$ and $R(2, 5, 7)$ meet the plane $3x + 2y + 11z - 9 = 0$ at Q. Then PQ is equal to :

(a) $\frac{5\sqrt{41}}{59}$ (b) $\frac{\sqrt{41}}{59}$ (c) $\frac{50\sqrt{41}}{59}$ (d) $\frac{25\sqrt{41}}{59}$

27. If $\vec{\mathbf{a}}, \vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ are three non-zero non-coplanar vectors and $\vec{\mathbf{p}} = \vec{\mathbf{a}} + \vec{\mathbf{b}} - 2\vec{\mathbf{c}}$; $\vec{\mathbf{q}} = 3\vec{\mathbf{a}} - 2\vec{\mathbf{b}} + \vec{\mathbf{c}}$ and $\vec{\mathbf{r}} = \vec{\mathbf{a}} - 4\vec{\mathbf{b}} + 2\vec{\mathbf{c}}$ are three vectors such that the volumes of the parallelopiped formed by $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ and $\vec{\mathbf{p}}, \vec{\mathbf{q}}, \vec{\mathbf{r}}$ as their coterminous edges are V_1 and V_2 respectively. Then $\frac{V_2}{V_1}$ is equal to :

(a) 10 (b) 15 (c) 20 (d) None of these

28. If the two lines represented by $x + ay = b$; $z + cy = d$ and $x = a'y + b'$; $z = c'y + d'$ be perpendicular to each other, then the value of $aa' + cc'$ is :

(a) 1 (b) 2 (c) 3 (d) 4

29. The distance between the line $\vec{\mathbf{r}} = 2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}} + \lambda(\hat{\mathbf{i}} - \hat{\mathbf{j}} + 4\hat{\mathbf{k}})$ and the plane $\vec{\mathbf{r}} \cdot (\hat{\mathbf{i}} + 5\hat{\mathbf{j}} + \hat{\mathbf{k}}) = 5$ is :

(a) $\frac{10}{9}$ (b) $\frac{10}{3\sqrt{3}}$ (c) $\frac{3}{10}$ (d) $\frac{10}{3}$

30. If $(\vec{\mathbf{a}} \times \vec{\mathbf{b}}) \times \vec{\mathbf{c}} = \vec{\mathbf{a}} \times (\vec{\mathbf{b}} \times \vec{\mathbf{c}})$, where $\vec{\mathbf{a}}, \vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ are any three vectors such that $\vec{\mathbf{a}} \cdot \vec{\mathbf{b}} \neq 0$, $\vec{\mathbf{b}} \cdot \vec{\mathbf{c}} \neq 0$, then $\vec{\mathbf{a}}$ and $\vec{\mathbf{c}}$ are :

(a) Inclined at an angle of $\frac{\pi}{3}$ (b) Inclined at an angle of $\frac{\pi}{6}$

(c) Perpendicular (d) Parallel

31. Let $\vec{\mathbf{r}}$ be position vector of variable point in cartesian plane OXY such that $\vec{\mathbf{r}}\cdot(\vec{\mathbf{r}}+6\hat{\mathbf{j}})=7$ cuts the co-ordinate axes at four distinct points, then the area of the quadrilateral formed by joining these points is :

(a) $4\sqrt{7}$ (b) $6\sqrt{7}$ (c) $7\sqrt{7}$ (d) $8\sqrt{7}$

32. If $|\vec{\mathbf{a}}|=2, |\vec{\mathbf{b}}|=5$ and $\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}=0$, then $\vec{\mathbf{a}}\times(\vec{\mathbf{a}}\times(\vec{\mathbf{a}}\times(\vec{\mathbf{a}}\times(\vec{\mathbf{a}}\times(\vec{\mathbf{a}}\times\vec{\mathbf{b}})))))$ is equal to :

(a) $64\vec{\mathbf{a}}$ (b) $64\vec{\mathbf{b}}$ (c) $-64\vec{\mathbf{a}}$ (d) $-64\vec{\mathbf{b}}$

33. If O (origin) is a point inside the triangle PQR such that $\vec{\mathbf{OP}}+k_1\vec{\mathbf{OQ}}+k_2\vec{\mathbf{OR}}=0$, where k_1, k_2 are constants such that $\dfrac{\text{Area }(\Delta PQR)}{\text{Area }(\Delta OQR)}=4$, then the value of k_1+k_2 is :

(a) 2 (b) 3 (c) 4 (d) 5

34. Let PQ and QR be diagonals of adjacent faces of a rectangular box, with its centre at O. If $\angle QOR$, $\angle ROP$ and $\angle POQ$ are θ, ϕ and Ψ respectively then the value of '$\cos\theta+\cos\phi+\cos\Psi$' is :

(a) -2 (b) $-\sqrt{3}$ (c) -1 (d) 0

35. The value of $\begin{vmatrix}\vec{\mathbf{a}} & \vec{\mathbf{b}} & \vec{\mathbf{c}}\\ \vec{\mathbf{a}}\cdot\vec{\mathbf{p}} & \vec{\mathbf{b}}\cdot\vec{\mathbf{p}} & \vec{\mathbf{c}}\cdot\vec{\mathbf{p}}\\ \vec{\mathbf{a}}\cdot\vec{\mathbf{q}} & \vec{\mathbf{b}}\cdot\vec{\mathbf{q}} & \vec{\mathbf{c}}\cdot\vec{\mathbf{q}}\end{vmatrix}$ is equal to :

(a) $(\vec{\mathbf{p}}\times\vec{\mathbf{q}})[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}\times\vec{\mathbf{c}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{a}}]$ (b) $2(\vec{\mathbf{p}}\times\vec{\mathbf{q}})[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}\times\vec{\mathbf{c}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{a}}]$

(c) $4(\vec{\mathbf{p}}\times\vec{\mathbf{q}})[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}\times\vec{\mathbf{c}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{a}}]$ (d) $(\vec{\mathbf{p}}\times\vec{\mathbf{q}})\sqrt{[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}\times\vec{\mathbf{c}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{a}}]}$

36. If $\vec{\mathbf{r}}=a(\vec{\mathbf{m}}\times\vec{\mathbf{n}})+b(\vec{\mathbf{n}}\times\vec{\mathbf{l}})+c(\vec{\mathbf{l}}\times\vec{\mathbf{m}})$ and $[\vec{\mathbf{l}}\,\vec{\mathbf{m}}\,\vec{\mathbf{n}}]=4$, find $\dfrac{a+b+c}{\vec{\mathbf{r}}\cdot(\vec{\mathbf{l}}+\vec{\mathbf{m}}+\vec{\mathbf{n}})}$:

(a) $\dfrac{1}{4}$ (b) $\dfrac{1}{2}$ (c) 1 (d) 2

37. The volume of tetrahedron, for which three co-terminus edges are $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$, is k units. Then, the volume of a parallelepiped formed by $\vec{\mathbf{a}}-\vec{\mathbf{b}}$, $\vec{\mathbf{b}}+2\vec{\mathbf{c}}$ and $3\vec{\mathbf{a}}-\vec{\mathbf{c}}$ is :

(a) $6k$ (b) $7k$ (c) $30k$ (d) $42k$

38. The equation of a plane passing through the line of intersection of the planes :
$x+2y+z-10=0$ and $3x+y-z=5$ and passing through the origin is :

(a) $5x+3z=0$ (b) $5x-3z=0$

(c) $5x+4y+3z=0$ (d) $5x-4y+3z=0$

39. Find the locus of a point whose distance from x -axis is twice the distance from the point $(1, -1, 2)$:

(a) $y^2 + 2x - 2y - 4z + 6 = 0$
(b) $x^2 + 2x - 2y - 4z + 6 = 0$
(c) $x^2 - 2x + 2y - 4z + 6 = 0$
(d) $z^2 - 2x + 2y - 4z + 6 = 0$

Answers

1.	(b)	**2.**	(b)	**3.**	(c)	**4.**	(b)	**5.**	(a)	**6.**	(c)	**7.**	(d)	**8.**	(c)	**9.**	(c)	**10.**	(d)
11.	(d)	**12.**	(d)	**13.**	(c)	**14.**	(c)	**15.**	(d)	**16.**	(b)	**17.**	(b)	**18.**	(b)	**19.**	(b)	**20.**	(b)
21.	(b)	**22.**	(a)	**23.**	(b)	**24.**	(a)	**25.**	(d)	**26.**	(d)	**27.**	(b)	**28.**	(a)	**29.**	(b)	**30.**	(d)
31.	(d)	**32.**	(d)	**33.**	(b)	**34.**	(c)	**35.**	(d)	**36.**	(a)	**37.**	(d)	**38.**	(b)	**39.**	(c)		

Exercise-2 : One or More than One Answer is/are Correct

1. If equation of three lines are :

$$\frac{x}{1}=\frac{y}{2}=\frac{z}{3};\ \frac{x}{2}=\frac{y}{1}=\frac{z}{3} \text{ and } \frac{x-1}{1}=\frac{2-y}{1}=\frac{z-3}{0}, \text{ then}$$

which of the following statement(s) is/are correct ?

(a) Triangle formed by the line is equilateral

(b) Triangle formed by the lines is isosceles

(c) Equation of the plane containing the lines is $x+y=z$

(d) Area of the triangle formed by the lines is $\frac{3\sqrt{3}}{2}$

2. If $\vec{\mathbf{a}}=\hat{\mathbf{i}}+6\hat{\mathbf{j}}+3\hat{\mathbf{k}}$; $\vec{\mathbf{b}}=3\hat{\mathbf{i}}+2\hat{\mathbf{j}}+\hat{\mathbf{k}}$ and $\vec{\mathbf{c}}=(\alpha+1)\hat{\mathbf{i}}+(\beta-1)\hat{\mathbf{j}}+\hat{\mathbf{k}}$ are linearly dependent vectors and $|\vec{\mathbf{c}}|=\sqrt{6}$; then the possible value(s) of $(\alpha+\beta)$ can be :

(a) 1 (b) 2 (c) 3 (d) 4

3. Consider the lines :

$$L_1 : \frac{x-2}{1}=\frac{y-1}{7}=\frac{z+2}{-5}$$

$$L_2 : x-4=y+3=-z$$

Then which of the following is/are correct ?

(a) Point of intersection of L_1 and L_2 is $(1,-6,3)$

(b) Equation of plane containing L_1 and L_2 is $x+2y+3z+2=0$

(c) Acute angle between L_1 and L_2 is $\cot^{-1}\left(\frac{13}{15}\right)$

(d) Equation of plane containing L_1 and L_2 is $x+2y+2z+3=0$

4. Let $\hat{\mathbf{a}}, \hat{\mathbf{b}}$ and $\hat{\mathbf{c}}$ be three unit vectors such that $\hat{\mathbf{a}}=\hat{\mathbf{b}}+(\hat{\mathbf{b}}\times\hat{\mathbf{c}})$, then the possible value(s) of $|\hat{\mathbf{a}}+\hat{\mathbf{b}}+\hat{\mathbf{c}}|^2$ can be :

(a) 1 (b) 4 (c) 16 (d) 9

5. The value(s) of μ for which the straight lines $\vec{\mathbf{r}}=3\hat{\mathbf{i}}-2\hat{\mathbf{j}}-4\hat{\mathbf{k}}+\lambda_1(\hat{\mathbf{i}}-\hat{\mathbf{j}}+\mu\hat{\mathbf{k}})$ and $\vec{\mathbf{r}}=5\hat{\mathbf{i}}-2\hat{\mathbf{j}}+\hat{\mathbf{k}}+\lambda_2(\hat{\mathbf{i}}+\mu\hat{\mathbf{j}}+2\hat{\mathbf{k}})$ are coplanar is/are :

(a) $\frac{5+\sqrt{33}}{4}$ (b) $\frac{-5+\sqrt{33}}{4}$ (c) $\frac{5-\sqrt{33}}{4}$ (d) $\frac{-5-\sqrt{33}}{4}$

6. If $\hat{\mathbf{i}}\times[(\vec{\mathbf{a}}-\hat{\mathbf{j}})\times\hat{\mathbf{i}}]+\hat{\mathbf{j}}\times[(\vec{\mathbf{a}}-\hat{\mathbf{k}})\times\hat{\mathbf{j}}]+\hat{\mathbf{k}}\times[(\vec{\mathbf{a}}-\hat{\mathbf{i}})\times\hat{\mathbf{k}}]=0$ and $\vec{\mathbf{a}}=x\hat{\mathbf{i}}+y\hat{\mathbf{j}}+z\hat{\mathbf{k}}$, then :

(a) $x+y=1$ (b) $y+z=\frac{1}{2}$ (c) $x+z=1$ (d) None of these

7. The value of expression $[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{d}}\ \ \vec{\mathbf{e}}\times\vec{\mathbf{f}}]$ is equal to :

(a) $[\vec{\mathbf{a}}\,\vec{\mathbf{b}}\,\vec{\mathbf{d}}][\vec{\mathbf{c}}\,\vec{\mathbf{e}}\,\vec{\mathbf{f}}]-[\vec{\mathbf{a}}\,\vec{\mathbf{b}}\,\vec{\mathbf{c}}][\vec{\mathbf{d}}\,\vec{\mathbf{e}}\,\vec{\mathbf{f}}]$

(b) $[\vec{\mathbf{a}}\,\vec{\mathbf{b}}\,\vec{\mathbf{e}}][\vec{\mathbf{f}}\,\vec{\mathbf{c}}\,\vec{\mathbf{d}}]-[\vec{\mathbf{a}}\,\vec{\mathbf{b}}\,\vec{\mathbf{f}}][\vec{\mathbf{e}}\,\vec{\mathbf{c}}\,\vec{\mathbf{d}}]$

(c) $[\vec{\mathbf{c}}\,\vec{\mathbf{d}}\,\vec{\mathbf{a}}][\vec{\mathbf{b}}\,\vec{\mathbf{e}}\,\vec{\mathbf{f}}]-[\vec{\mathbf{c}}\,\vec{\mathbf{d}}\,\vec{\mathbf{b}}][\vec{\mathbf{a}}\,\vec{\mathbf{e}}\,\vec{\mathbf{f}}]$

(d) $[\vec{\mathbf{b}}\,\vec{\mathbf{c}}\,\vec{\mathbf{d}}][\vec{\mathbf{a}}\,\vec{\mathbf{e}}\,\vec{\mathbf{f}}]-[\vec{\mathbf{b}}\,\vec{\mathbf{c}}\,\vec{\mathbf{f}}][\vec{\mathbf{a}}\,\vec{\mathbf{e}}\,\vec{\mathbf{d}}]$

8. If $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ and $\vec{\mathbf{d}}$ are the position vectors of the points A, B, C and D respectively in three dimensional space and satisfy the relation $3\vec{\mathbf{a}}-2\vec{\mathbf{b}}+\vec{\mathbf{c}}-2\vec{\mathbf{d}}=0$, then :

(a) A, B, C and D are coplanar

(b) The line joining the points B and D divides the line joining the point A and C in the ratio of $2:1$

(c) The line joining the points A and C divides the line joining the points B and D in the ratio of $1:1$

(d) The four vectors $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ and $\vec{\mathbf{d}}$ are linearly dependent.

9. If $OABC$ is a tetrahedron with equal edges and $\hat{\mathbf{p}}, \hat{\mathbf{q}}, \hat{\mathbf{r}}$ are unit vectors along bisectors of $\overrightarrow{\mathbf{OA}}, \overrightarrow{\mathbf{OB}} : \overrightarrow{\mathbf{OB}}, \overrightarrow{\mathbf{OC}} : \overrightarrow{\mathbf{OC}}, \overrightarrow{\mathbf{OA}}$ respectively and $\hat{\mathbf{a}}=\dfrac{\overrightarrow{\mathbf{OA}}}{|\overrightarrow{\mathbf{OA}}|}, \vec{\mathbf{b}}=\dfrac{\overrightarrow{\mathbf{OB}}}{|\overrightarrow{\mathbf{OB}}|}, \vec{\mathbf{c}}=\dfrac{\overrightarrow{\mathbf{OC}}}{|\overrightarrow{\mathbf{OC}}|}$, then :

(a) $\dfrac{[\hat{\mathbf{a}}\,\hat{\mathbf{b}}\,\hat{\mathbf{c}}]}{[\hat{\mathbf{p}}\,\hat{\mathbf{q}}\,\hat{\mathbf{r}}]}=\dfrac{3\sqrt{3}}{2}$

(b) $\dfrac{[\hat{\mathbf{a}}+\hat{\mathbf{b}}\ \ \hat{\mathbf{b}}+\hat{\mathbf{c}}\ \ \hat{\mathbf{c}}+\hat{\mathbf{a}}]}{[\hat{\mathbf{p}}+\hat{\mathbf{q}}\ \ \hat{\mathbf{q}}+\hat{\mathbf{r}}\ \ \hat{\mathbf{r}}+\hat{\mathbf{p}}]}=\dfrac{3\sqrt{3}}{4}$

(c) $\dfrac{[\hat{\mathbf{a}}+\hat{\mathbf{b}}\ \ \hat{\mathbf{b}}+\hat{\mathbf{c}}\ \ \hat{\mathbf{c}}+\hat{\mathbf{a}}]}{[\hat{\mathbf{p}}\,\hat{\mathbf{q}}\,\hat{\mathbf{r}}]}=\dfrac{3\sqrt{3}}{2}$

(d) $\dfrac{[\hat{\mathbf{a}}\,\hat{\mathbf{b}}\,\hat{\mathbf{c}}]}{[\hat{\mathbf{p}}+\hat{\mathbf{q}}\ \ \hat{\mathbf{q}}+\hat{\mathbf{r}}\ \ \hat{\mathbf{r}}+\hat{\mathbf{p}}]}=\dfrac{3\sqrt{3}}{4}$

10. Let $\hat{\mathbf{a}}$ and $\hat{\mathbf{c}}$ are unit vectors and $|\vec{\mathbf{b}}|=4$. If the angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{c}}$ is $\cos^{-1}\left(\dfrac{1}{4}\right)$; and $\vec{\mathbf{b}}-2\hat{\mathbf{c}}=\lambda\hat{\mathbf{a}}$, then the value of λ can be :

(a) 2

(b) –3

(c) 3

(d) –4

11. Consider the line $L_1: x=y=z$ and the line $L_2: 2x+y+z-1=0=3x+y+2z-2$, then :

(a) The shortest distance between the two lines is $\dfrac{1}{\sqrt{2}}$

(b) The shortest distance between the two lines is $\sqrt{2}$

(c) Plane containing the line L_2 and parallel to line L_1 is $z-x+1=0$

(d) Perpendicular distance of origin from plane containing line L_2 and parallel to line L_1 is $\dfrac{1}{\sqrt{2}}$

12. Let $\vec{r} = \sin x(\vec{a} \times \vec{b}) + \cos y(\vec{b} \times \vec{c}) + 2(\vec{c} \times \vec{a})$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar vectors. It is given that $\vec{r}$ is perpendicular to $\vec{a} + \vec{b} + \vec{c}$. The possible value(s) of $x^2 + y^2$ is/are :

(a) π^2 (b) $\dfrac{5\pi^2}{4}$

(c) $\dfrac{35\pi^2}{4}$ (d) $\dfrac{37\pi^2}{4}$

13. If $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = h\,\vec{a} + k\,\vec{b} = r\,\vec{c} + s\,\vec{d}$, where $\vec{a}$, $\vec{b}$ are non-collinear and $\vec{c}, \vec{d}$ are also non-collinear then :

(a) $h = [\vec{b}\ \vec{c}\ \vec{d}]$ (b) $k = [\vec{a}\ \vec{c}\ \vec{d}]$

(c) $r = [\vec{a}\ \vec{b}\ \vec{d}]$ (d) $s = -[\vec{a}\ \vec{b}\ \vec{c}]$

14. Let a be a real number and $\vec{\alpha} = \hat{\mathbf{i}} + 2\hat{\mathbf{j}}$, $\vec{\beta} = 2\hat{\mathbf{i}} + a\hat{\mathbf{j}} + 10\hat{\mathbf{k}}$, $\vec{\gamma} = 12\hat{\mathbf{i}} + 20\hat{\mathbf{j}} + a\hat{\mathbf{k}}$ be three vectors, then $\vec{\alpha}, \vec{\beta}$ and $\vec{\gamma}$ are linearly independent for :

(a) $a > 0$ (b) $a < 0$

(c) $a = 0$ (d) No value of a

15. The volume of a right triangular prism $ABCA_1B_1C_1$ is equal to 3. If the position vectors of the vertices of the base ABC are $A(1, 0, 1)$; $B(2, 0, 0)$ and $C(0, 1, 0)$, then the position vectors of the vertex A_1 can be :

(a) (2, 2, 2) (b) (0, 2, 0)

(c) (0, –2, 2) (d) (0, –2, 0)

16. If $\vec{a} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, $\vec{b} = y\hat{\mathbf{i}} + z\hat{\mathbf{j}} + x\hat{\mathbf{k}}$, and $\vec{c} = z\hat{\mathbf{i}} + x\hat{\mathbf{j}} + y\hat{\mathbf{k}}$, then $\vec{a} \times (\vec{b} \times \vec{c})$ is :

(a) Parallel to $(y - z)\hat{\mathbf{i}} + (z - x)\hat{\mathbf{j}} + (x - y)\hat{\mathbf{k}}$

(b) Orthogonal to $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$

(c) Orthogonal to $(y + z)\hat{\mathbf{i}} + (z + x)\hat{\mathbf{j}} + (x + y)\hat{\mathbf{k}}$,

(d) Orthogonal to $x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$

17. If a line has a vector equation, $\vec{r} = 2\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + \lambda(\hat{\mathbf{i}} - 3\hat{\mathbf{j}})$ then which of the following statements holds good ?

(a) the line is parallel to $2\hat{\mathbf{i}} + 6\hat{\mathbf{j}}$

(b) the line passes through the point $3\hat{\mathbf{i}} + 3\hat{\mathbf{j}}$

(c) the line passes through the point $\hat{\mathbf{i}} + 9\hat{\mathbf{j}}$

(d) the line is parallel to xy plane

18. Let M, N, P and Q be the mid points of the edges AB, CD, AC and BD respectively of the tetrahedron $ABCD$. Further, MN is perpendicular to both AB and CD and PQ is perpendicular to both AC and BD. Then which of the following is/are correct :

(a) $AB = CD$
(b) $BC = DA$
(c) $AC = BD$
(d) $AN = BN$

19. The solution vectors $\vec{\mathbf{r}}$ of the equation $\vec{\mathbf{r}} \times \hat{\mathbf{i}} = \hat{\mathbf{j}} + \hat{\mathbf{k}}$ and $\vec{\mathbf{r}} \times \hat{\mathbf{j}} = \hat{\mathbf{k}} + \hat{\mathbf{i}}$ represent two straight lines which are :

(a) Intersecting
(b) Non coplanar
(c) Coplanar
(d) Non intersecting

20. Which of the following statement(s) is/are incorrect ?

(a) The lines $\frac{x-4}{-3} = \frac{y+6}{-1} = \frac{z+6}{-1}$ and $\frac{x-1}{-1} = \frac{y-2}{-2} = \frac{z-3}{2}$ are orthogonal

(b) The planes $3x - 2y - 4z = 3$ and the plane $x - y - z = 3$ are orthogonal

(c) The function $f(x) = ln(e^{-2} + e^{x})$ is monotonic increasing $\forall \ x \in R$

(d) If g is the inverse of the function, $f(x) = \ln(e^{-2} + e^{x})$ then $g(x) = \ln(e^{x} - e^{-2})$

21. The lines with vector equations are; $\vec{\mathbf{r}}_1 = -3\hat{\mathbf{i}} + 6\hat{\mathbf{j}} + \lambda(-4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$ and $\vec{\mathbf{r}}_2 = -2\hat{\mathbf{i}} + 7\hat{\mathbf{j}} + \mu(-4\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}})$ are such that :

(a) they are coplanar
(b) they do not intersect
(c) they are skew
(d) the angle between them is $\tan^{-1}(3/7)$

Answers

1.	(b, c, d)	**2.**	(a, c)	**3.**	(a, b, c)	**4.**	(a, d)	**5.**	(a, c)	**6.**	(a, c)
7.	(a, b, c)	**8.**	(a, c, d)	**9.**	(a, d)	**10.**	(c, d)	**11.**	(a, d)	**12.**	(b, d)
13.	(b, c, d)	**14.**	(a, b, c)	**15.**	(a, d)	**16.**	(a, b, c, d)	**17.**	(b, c, d)	**18.**	(a, b, c, d)
19.	(b, d)	**20.**	(a, b)	**21.**	(b, c, d)						

Exercise-3 : Comprehension Type Problems

Paragraph for Question Nos. 1 to 3

The vertices of ΔABC are $A(2, 0, 0), B(0, 1, 0), C(0, 0, 2)$. Its orthocentre is H and circumcentre is S. P is a point equidistant from A, B, C and the origin O.

1. The z-coordinate of H is :

(a) 1 (b) 1/2 (c) 1/6 (d) 1/3

2. The y-coordinate of S is :

(a) 5/6 (b) 1/3 (c) 1/6 (d) 1/2

3. PA is equal to :

(a) 1 (b) $\sqrt{2}$ (c) $\sqrt{\frac{3}{2}}$ (d) $\frac{3}{2}$

Paragraph for Question Nos. 4 to 6

Consider a plane $\pi: \vec{\mathbf{r}} \cdot \vec{\mathbf{n}} = d$ (where $\vec{\mathbf{n}}$ is not a unit vector). There are two points $A(\vec{\mathbf{a}})$ and $B(\vec{\mathbf{b}})$ lying on the same side of the plane.

4. If foot of perpendicular from A and B to the plane π are P and Q respectively, then length of PQ be:

(a) $\dfrac{|(\vec{\mathbf{b}}-\vec{\mathbf{a}}) \cdot \vec{\mathbf{n}}|}{|\vec{\mathbf{n}}|}$ (b) $|(\vec{\mathbf{b}}-\vec{\mathbf{a}}) \cdot \vec{\mathbf{n}}|$ (c) $\dfrac{|(\vec{\mathbf{b}}-\vec{\mathbf{a}}) \times \vec{\mathbf{n}}|}{|\vec{\mathbf{n}}|}$ (d) $|(\vec{\mathbf{b}}-\vec{\mathbf{a}}) \times \vec{\mathbf{n}}|$

5. Reflection of $A(\vec{\mathbf{a}})$ in the plane π has the position vector :

(a) $\vec{\mathbf{a}} + \dfrac{2}{(\vec{\mathbf{n}})^2}(d - \vec{\mathbf{a}} \cdot \vec{\mathbf{n}})\vec{\mathbf{n}}$ (b) $\vec{\mathbf{a}} - \dfrac{1}{(\vec{\mathbf{n}})^2}(d - \vec{\mathbf{a}} \cdot \vec{\mathbf{n}})\vec{\mathbf{n}}$

(c) $\vec{\mathbf{a}} + \dfrac{2}{(\vec{\mathbf{n}})^2}(d + \vec{\mathbf{a}} \cdot \vec{\mathbf{n}})\vec{\mathbf{n}}$ (d) $\vec{\mathbf{a}} + \dfrac{2}{(\vec{\mathbf{n}})^2}\vec{\mathbf{n}}$

6. If a plane π_1 is drawn from the point $A(\vec{\mathbf{a}})$ and another plane π_2 is drawn from point $B(\vec{\mathbf{b}})$ parallel to π, then the distance between the planes π_1 and π_2 is :

(a) $\dfrac{|(\vec{\mathbf{a}}-\vec{\mathbf{b}}) \cdot \vec{\mathbf{n}}|}{|\vec{\mathbf{n}}|}$ (b) $|(\vec{\mathbf{a}}-\vec{\mathbf{b}}) \cdot \vec{\mathbf{n}}|$ (c) $|(\vec{\mathbf{a}}-\vec{\mathbf{b}}) \times \vec{\mathbf{n}}|$ (d) $\dfrac{|(\vec{\mathbf{a}}-\vec{\mathbf{b}}) \times \vec{\mathbf{n}}|}{|\vec{\mathbf{n}}|}$

Paragraph for Question Nos. 7 to 9

Consider a plane $\prod : \vec{\mathbf{r}} \cdot (2\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) = 5$, a line $L_1 : \vec{\mathbf{r}} = (3\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}) + \lambda(2\hat{\mathbf{i}} - 3\hat{\mathbf{j}} - \hat{\mathbf{k}})$ and a point $A(3, -4, 1)$. L_2 is a line passing through A intersecting L_1 and parallel to plane Π.

7. Equation of L_2 is :

(a) $\vec{\mathbf{r}} = (1 + \lambda)\hat{\mathbf{i}} + (2 - 3\lambda)\hat{\mathbf{j}} + (1 - \lambda)\hat{\mathbf{k}}; \lambda \in R$

(b) $\vec{\mathbf{r}} = (3 + \lambda)\hat{\mathbf{i}} - (4 - 2\lambda)\hat{\mathbf{j}} + (1 + 3\lambda)\hat{\mathbf{k}}; \lambda \in R$

(c) $\vec{\mathbf{r}} = (3 + \lambda)\hat{\mathbf{i}} - (4 + 3\lambda)\hat{\mathbf{j}} + (1 - \lambda)\hat{\mathbf{k}}; \lambda \in R$

(d) None of the above

8. Plane containing L_1 and L_2 is :

(a) parallel to yz-plane
(b) parallel to x-axis
(c) parallel to y-axis
(d) passing through origin

9. Line L_1 intersects plane Π at Q and xy-plane at R the volume of tetrahedron $OAQR$ is : (where 'O' is origin)

(a) 0 (b) $\frac{14}{3}$ (c) $\frac{3}{7}$ (d) $\frac{7}{3}$

Paragraph for Question Nos. 10 to 11

Consider three planes :

$$2x + py + 6z = 8; x + 2y + qz = 5 \text{ and } x + y + 3z = 4$$

10. Three planes intersect at a point if :

(a) $p = 2, q \neq 3$ (b) $p \neq 2, q \neq 3$ (c) $p \neq 2, q = 3$ (d) $p = 2, q = 3$

11. Three planes do not have any common point of intersection if :

(a) $p = 2, q \neq 3$ (b) $p \neq 2, q \neq 3$ (c) $p \neq 2, q = 3$ (d) $p = 2, q = 3$

Paragraph for Question Nos. 12 to 14

The points A, B and C with position vectors $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ respectively lie on a circle centered at origin O. Let G and E be the centroid of ΔABC and ΔACD respectively where D is mid point of AB.

12. If OE and CD are mutually perpendicular, then which of the following will be necessarily true ?

(a) $|\vec{\mathbf{b}} - \vec{\mathbf{a}}| = |\vec{\mathbf{c}} - \vec{\mathbf{a}}|$
(b) $|\vec{\mathbf{b}} - \vec{\mathbf{a}}| = |\vec{\mathbf{b}} - \vec{\mathbf{c}}|$
(c) $|\vec{\mathbf{c}} - \vec{\mathbf{a}}| = |\vec{\mathbf{c}} - \vec{\mathbf{b}}|$
(d) $|\vec{\mathbf{b}} - \vec{\mathbf{a}}| = |\vec{\mathbf{c}} - \vec{\mathbf{a}}| = |\vec{\mathbf{b}} - \vec{\mathbf{c}}|$

13. If GE and CD are mutually perpendicular, then orthocenter of ΔABC must lie on :

(a) median through A (b) median through C

(c) angle bisector through A (d) angle bisector through B

14. If $[\overrightarrow{\mathbf{AB}}\ \overrightarrow{\mathbf{AC}}\ \overrightarrow{\mathbf{AB}} \times \overrightarrow{\mathbf{AC}}] = \lambda [\overrightarrow{\mathbf{AE}}\ \overrightarrow{\mathbf{AG}}\ \overrightarrow{\mathbf{AE}} \times \overrightarrow{\mathbf{AG}}]$, then the value of λ is :

(a) –18 (b) 18 (c) –324 (d) 324

Paragraph for Question Nos. 15 to 16

Consider a tetrahedron D —ABC with position vectors if its angular points as

$$A\ (1, 1, 1);\ B(1, 2, 3)\ ;\ C(1, 1, 2)$$

and centre of tetrahedron $\left(\frac{3}{2}, \frac{3}{4}, 2\right)$.

15. Shortest distance between the skew lines AB and CD :

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$

16. If N be the foot of the perpendicular from point D on the plane face ABC then the position vector of N are :

(a) (–1, 1, 2) (b) (1, –1, 2) (c) (1, 1, –2) (d) (–1, –1, 2)

Paragraph for Question Nos. 17 to 18

In a triangle AOB, R and Q are the points on the side OB and AB respectively such that $3OR = 2RB$ and $2AQ = 3QB$. Let OQ and AR intersect at the point P (where O is origin).

17. If the point P divides OQ in the ratio of $\mu : 1$, then μ is :

(a) $\frac{2}{19}$ (b) $\frac{2}{17}$ (c) $\frac{2}{15}$ (d) $\frac{10}{9}$

18. If the ratio of area of quadrilateral $PQBR$ and area of ΔOPA is $\frac{\alpha}{\beta}$ then $(\beta - \alpha)$ is (where α and β are coprime numbers) :

(a) 1 (b) 9 (c) 7 (d) 0

Answers

1.	(d)	**2.**	(c)	**3.**	(d)	**4.**	(c)	**5.**	(a)	**6.**	(a)	**7.**	(c)	**8.**	(b)	**9.**	(d)	**10.**	(b)
11.	(c)	**12.**	(a)	**13.**	(b)	**14.**	(d)	**15.**	(b)	**16.**	(b)	**17.**	(d)	**18.**	(d)				

Exercise-4 : Matching Type Problems

1.

	Column-I		Column-II
(A)	Lines $\dfrac{x-1}{-2}=\dfrac{y+2}{3}=\dfrac{z}{-1}$ and $\vec{\mathbf{r}}=(3\hat{i}-\hat{j}+\hat{k})+t(\hat{i}+\hat{j}+\hat{k})$ are	**(P)**	Intersecting
(B)	Lines $\dfrac{x+5}{1}=\dfrac{y-3}{7}=\dfrac{z+3}{3}$ and $x-y+2z-4=0=2x+y-3z+5$ are	**(Q)**	Perpendicular
(C)	Lines $(x=t-3,\ y=-2t+1,\ z=-3t-2)$ and $\vec{\mathbf{r}}=(t+1)\hat{i}+(2t+3)\hat{j}+(-t-9)\hat{k}$ are	**(R)**	Parallel
(D)	Lines $\vec{\mathbf{r}}=(\hat{i}+3\hat{j}-\hat{k})+t\,(2\hat{i}-\hat{j}-\hat{k})$ and $\vec{\mathbf{r}}=(-\hat{i}-2\hat{j}+5\hat{k})+s\left(\hat{i}-2\hat{j}+\dfrac{3}{4}\hat{k}\right)$ are	**(S)**	Skew
		(T)	Coincident

2.

	Column-I		Column-II
(A)	If $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ are three mutually perpendicular vectors where $\lvert\vec{\mathbf{a}}\rvert=\lvert\vec{\mathbf{b}}\rvert=2$, $\lvert\vec{\mathbf{c}}\rvert=1$, then $[\vec{\mathbf{a}}\times\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}\times\vec{\mathbf{c}}\ \ \vec{\mathbf{c}}\times\vec{\mathbf{a}}]$ is	**(P)**	-12
(B)	If $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are two unit vectors inclined at $\dfrac{\pi}{3}$, then $16[\vec{\mathbf{a}}\ \ \vec{\mathbf{b}}+(\vec{\mathbf{a}}\times\vec{\mathbf{b}})\ \ \vec{\mathbf{b}}]$ is	**(Q)**	0
(C)	If $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}}$ are orthogonal unit vectors and $\vec{\mathbf{b}}\times\vec{\mathbf{c}}=\vec{\mathbf{a}}$ then $[\vec{\mathbf{a}}+\vec{\mathbf{b}}+\vec{\mathbf{c}}\ \ \vec{\mathbf{a}}+\vec{\mathbf{b}}\ \ \vec{\mathbf{b}}+\vec{\mathbf{c}}]$ is	**(R)**	16
(D)	If $[\vec{\mathbf{x}}\ \vec{\mathbf{y}}\ \vec{\mathbf{a}}]=[\vec{\mathbf{x}}\ \vec{\mathbf{y}}\ \vec{\mathbf{b}}]=[\vec{\mathbf{a}}\ \vec{\mathbf{b}}\ \vec{\mathbf{c}}]=0$, each vector being a non-zero vector, then $[\vec{\mathbf{x}}\ \vec{\mathbf{y}}\ \vec{\mathbf{c}}]$ is	**(S)**	1
		(T)	4

3.

	Column-I		Column-II
(A)	The number of real roots of equation $2^x + 3^x + 4^x - 9^x = 0$ is λ, then $\lambda^2 + 7$ is divisible by	**(P)**	2
(B)	Let ABC be a triangle whose centroid is G, orthocenter is H and circumcentre is the origin 'O'. If D is any point in the plane of the triangle such that not three of O, A, B, C and D are collinear satisfying the relation $\overrightarrow{AD} + \overrightarrow{BD} + \overrightarrow{CH} + 3\overrightarrow{HG} = \lambda\overrightarrow{HD}$, then $\lambda + 4$ is divisible by	**(Q)**	3
(C)	If $A\ (\text{adj } A) = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$, then $5\lvert A\rvert - 2$ is divisible by	**(R)**	4
(D)	$\vec{a}, \vec{b}, \vec{c}$ are three unit vector such that $\vec{a} + \vec{b} = \sqrt{2}\,\vec{c}$, then $\lvert 6\vec{a} - 8\vec{b}\rvert$ is divisible by	**(S)**	6
		(T)	10

Answers

1. A $\rightarrow$ Q, S; B $\rightarrow$ R; C $\rightarrow$ P, Q; D $\rightarrow$ P

2. A $\rightarrow$ R; B $\rightarrow$ P; C $\rightarrow$ S; D $\rightarrow$ Q

3. A $\rightarrow$ P, R; B $\rightarrow$ P, Q, S ; C $\rightarrow$ P, Q, R, S; D $\rightarrow$ P, T

Exercise-5 : Subjective Type Problems

1. A straight line L intersects perpendicularly both the lines :

$$\frac{x+2}{2}=\frac{y+6}{3}=\frac{z-34}{-10} \text{ and } \frac{x+6}{4}=\frac{y-7}{-3}=\frac{z-7}{-2},$$

then the square of perpendicular distance of origin from L is

2. If $\hat{\mathbf{a}}, \hat{\mathbf{b}}$ and $\hat{\mathbf{c}}$ are non-coplanar unit vectors such that $[\hat{\mathbf{a}}\,\hat{\mathbf{b}}\,\hat{\mathbf{c}}]=[\hat{\mathbf{b}}\times\hat{\mathbf{c}}\ \ \hat{\mathbf{c}}\times\hat{\mathbf{a}}\ \ \hat{\mathbf{a}}\times\hat{\mathbf{b}}]$, then find the projection of $\hat{\mathbf{b}}+\hat{\mathbf{c}}$ on $\hat{\mathbf{a}}\times\hat{\mathbf{b}}$.

3. Let OA, OB, OC be coterminous edges of a cuboid. If l, m, n be the shortest distances between the sides OA, OB, OC and their respective skew body diagonals to them, respectively, then find

$$\frac{\left(\frac{1}{l^2}+\frac{1}{m^2}+\frac{1}{n^2}\right)}{\left(\frac{1}{OA^2}+\frac{1}{OB^2}+\frac{1}{OC^2}\right)}.$$

4. Let $OABC$ be a tetrahedron whose edges are of unit length. If $\overrightarrow{\mathbf{OA}}=\vec{\mathbf{a}}$, $\overrightarrow{\mathbf{OB}}=\vec{\mathbf{b}}$ and d $\overrightarrow{\mathbf{OC}}=\alpha(\vec{\mathbf{a}}+\vec{\mathbf{b}})+\beta(\vec{\mathbf{a}}\times\vec{\mathbf{b}})$, then $(\alpha\beta)^2=\frac{p}{q}$ where p and q are relatively prime to each other. Find the value of $\left[\frac{q}{2p}\right]$ where [·] denotes greatest integer function.

5. Let $\vec{\mathbf{v}}_0$ be a fixed vector and $\vec{\mathbf{v}}_0=\begin{bmatrix}1\\0\end{bmatrix}$. Then for $n\geq 0$ a sequence is defined $\vec{\mathbf{v}}_{n+1}=\vec{\mathbf{v}}_n+\left(\frac{1}{2}\right)^{n+1}\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}^{n+1}\vec{\mathbf{v}}_0$ then $\lim\limits_{n\to\infty}\vec{\mathbf{v}}_n=\begin{bmatrix}\alpha\\\beta\end{bmatrix}$. Find $\frac{\alpha}{\beta}$.

6. If A is the matrix $\begin{bmatrix}1 & -3\\-1 & 1\end{bmatrix}$, then $A-\frac{1}{3}A^2+\frac{1}{9}A^3\ldots\ldots+\left(-\frac{1}{3}\right)^n A^{n+1}+\ldots\ldots\infty=\frac{3}{13}\begin{bmatrix}1 & a\\b & 1\end{bmatrix}$. Find $\left|\frac{a}{b}\right|$.

7. A sequence of 2×2 matrices $\{M_n\}$ is defined as follows $M_n=\begin{bmatrix}\dfrac{1}{(2n+1)!} & \dfrac{1}{(2n+2)!}\\ \sum\limits_{k=0}^{n}\dfrac{(2n+2)!}{(2k+2)!} & \sum\limits_{k=0}^{n}\dfrac{(2n+1)!}{(2k+1)!}\end{bmatrix}$

then $\lim\limits_{n\to\infty}\det.(M_n)=\lambda-e^{-1}$. Find λ.

8. Let $|\vec{\mathbf{a}}|=1$, $|\vec{\mathbf{b}}|=1$ and $|\vec{\mathbf{a}}+\vec{\mathbf{b}}|=\sqrt{3}$. If $\vec{\mathbf{c}}$ be a vector such that $\vec{\mathbf{c}}=\vec{\mathbf{a}}+2\vec{\mathbf{b}}-3(\vec{\mathbf{a}}\times\vec{\mathbf{b}})$ and $p=|(\vec{\mathbf{a}}\times\vec{\mathbf{b}})\times\vec{\mathbf{c}}|$, then find $[p^2]$. (where [ ] represents greatest integer function).

9. Let $\vec{\mathbf{r}} = (\vec{\mathbf{a}} \times \vec{\mathbf{b}})\sin x + (\vec{\mathbf{b}} \times \vec{\mathbf{c}})\cos y + 2(\vec{\mathbf{c}} \times \vec{\mathbf{a}})$, where $\vec{\mathbf{a}}, \vec{\mathbf{b}}, \vec{\mathbf{c}}$ are non-zero and non-coplanar vectors. If $\vec{\mathbf{r}}$ is orthogonal to $\vec{\mathbf{a}} + \vec{\mathbf{b}} + \vec{\mathbf{c}}$, then find the minimum value of $\frac{4}{\pi^2}(x^2 + y^2)$.

10. The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = 25$. If the plane in its new position be denoted by Π, and the distance of this plane from the origin is $\sqrt{53\,k}$ where $k \in N$, then find k.

11. $ABCD$ is a regular tetrahedron, A is the origin and B lies on x-axis. ABC lies in the xy-plane $|\vec{\mathbf{AB}}| = 2$. Under these conditions, the number of possible tetrahedrons is :

12. A, B, C, D are four points in the space and satisfy $|\vec{\mathbf{AB}}| = 3$, $|\vec{\mathbf{BC}}| = 7$, $|\vec{\mathbf{CD}}| = 11$ and $|\vec{\mathbf{DA}}| = 9$. Then find the value of $\vec{\mathbf{AC}} \cdot \vec{\mathbf{BD}}$.

13. Let $OABC$ be a regular tetrahedron of edge length unity. Its volume be V and $6V = \sqrt{p/q}$ where p and q are relatively prime. The find the value of $(p + q)$:

14. If $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are non zero, non collinear vectors and $\vec{\mathbf{a}}_1 = \lambda\vec{\mathbf{a}} + 3\vec{\mathbf{b}}$; $\vec{\mathbf{b}}_1 = 2\vec{\mathbf{a}} + \lambda\vec{\mathbf{b}}$; $\vec{\mathbf{c}}_1 = \vec{\mathbf{a}} + \vec{\mathbf{b}}$. Find the sum of all possible real values of λ so that points A_1, B_1, C_1 whose position vectors are $\vec{\mathbf{a}}_1, \vec{\mathbf{b}}_1, \vec{\mathbf{c}}_1$ respectively are collinear is equal to .

15. Let P and Q are two points on curve $y = \log_{\frac{1}{2}}\left(x - \frac{1}{2}\right) + \log_2\sqrt{4x^2 - 4x + 1}$ and P is also on $x^2 + y^2 = 10$. Q lies inside the given circle such that its abscissa is integer. Find the smallest possible value of $\vec{\mathbf{OP}} \cdot \vec{\mathbf{OQ}}$ where 'O' being origin.

16. In above problem find the largest possible value of $|\vec{\mathbf{PQ}}|$.

17. If $a, b, c, l, m, n \in R - \{0\}$ such that $al + bm + cn = 0$, $bl + cm + an = 0$, $cl + am + bn = 0$. If a, b, c are distinct and $f(x) = ax^3 + bx^2 + cx + 2$. Find $f(1)$:

18. Let $\vec{\mu}$ and $\vec{\nu}$ are unit vectors and $\vec{\omega}$ is vector such that $\vec{\mu} \times \vec{\nu} + \vec{\mu} = \vec{\omega}$ and $\vec{\omega} \times \vec{\mu} = \vec{\nu}$. The find the value of $[\vec{\mu}\ \vec{\nu}\ \vec{\omega}]$.

Answers

1.	5	**2.**	1	**3.**	2	**4.**	5	**5.**	2	**6.**	3	**7.**	1
8.	5	**9.**	5	**10.**	4	**11.**	8	**12.**	0	**13.**	0	**14.**	2
15.	4	**16.**	2	**17.**	2	**18.**	1						

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